BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 5 Packet.

Urinary · Reproductive · Brain · Spinal Cord · Peripheral · Autonomic

THIS PACKET	**Course Notes**	13 chapters, one section per competency, with the competency number on every section. Contents at the front, index at the back.
SEPARATE	**Lab Packet**	The lab structure list, muscle and flow charts, and the questions you could be asked on the lab exam.
SEPARATE	**Note Sheets**	One drawing sheet per competency, two prompts each, with an answer checklist at the back.

BIO 004 · MODULE 5

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Every chapter and every competency, with its page. The lab packet and the note sheets are separate documents.

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Everything else for Module 5

Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

WHAT MODULE 5 COVERS

Module 5 is the last module, and the final exam closes it. It is sixty one competencies, numbers 133 to 193 in your competency packet. Every section of these notes is headed with its competency number, so you can go from the packet to the page and back.

It opens with the urinary and reproductive systems, which finish the organ systems. The rest of the module is the nervous system: the brain and brainstem, the meninges and cerebrospinal fluid, the tissue and wiring logic, the spinal cord, and the nerves that run out to the body.

Most nervous system questions are pathway questions or lesion questions, and both need the wiring in your head before the names will stick.

CHAPTER 1

The Urinary System

URINARY

COMPETENCIES 133 TO 140

Covers the gross anatomy of the kidney, the nephron, the juxtaglomerular apparatus and filtration membrane, the blood supply, and the ureters, bladder, and urethra. The focus is on the structures and the job each one does.

COMPETENCIES

1. **133** Kidney gross anatomy and layers
2. **134** Internal kidney regions
3. **135** Nephron structure
4. **136** Cortical vs juxtamedullary nephrons
5. **137** JGA and filtration membrane
6. **138** Kidney blood supply pathway
7. **139** Ureters, bladder, urethra and urine path
8. **140** Clinical anatomy of urinary disorders

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The urinary system

drsrennie-stack.github.io/new-build-bio4-solano/urinary-concept-videos.html

COMPETENCY 133 LECTURE AND LAB

Kidney gross anatomy and layers

You should be able to: Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia.

The **urinary system** is made of two kidneys, two ureters, the urinary bladder, and the urethra. The kidneys filter the blood and make urine. The rest of the system carries the urine away, stores it, and releases it.

THE PARTS OF THE URINARY SYSTEM

Structure	What it is
Kidneys	The paired organs that filter the blood and form urine
Ureters	The paired tubes that carry urine from the kidneys to the bladder
Urinary bladder	The muscular sac that stores urine
Urethra	The tube that carries urine from the bladder to the outside
Roles of the kidney	Filtering wastes from the blood, balancing water, ions, and pH, and helping regulate blood pressure

Where the kidneys sit

The **kidneys** are bean-shaped organs on the posterior abdominal wall, one on each side of the vertebral column. They lie roughly between the T12 and L3 vertebrae. The lowest ribs partly cover their upper ends.

The kidneys are **retroperitoneal**. This means they sit behind the parietal peritoneum, outside the peritoneal cavity, against the muscles of the back.

The **right kidney** sits a little lower than the **left kidney** because the liver takes up the space above it. A **suprarenal (adrenal) gland** sits on top of each kidney like a cap. The adrenal gland is part of the endocrine system, but the same

renal fascia wraps it together with the kidney.

The hilum

Each kidney has a rounded lateral border and a concave medial border. The notch in the medial border is the **renal hilum**. The renal artery, renal vein, nerves, lymphatic vessels, and ureter all enter or leave the kidney here. Just inside the hilum is the **renal sinus**, a space filled with fat that holds the renal pelvis, the calyces, and the branching vessels.

The three tissue layers

Three layers of tissue surround each kidney. From the kidney outward they are the renal capsule, the adipose capsule, and the renal fascia. Each layer has its own job.

THE THREE LAYERS AROUND THE KIDNEY, FROM THE INSIDE OUT

Layer	Tissue	Job
Renal capsule (innermost)	A thin, tough layer of dense fibrous connective tissue on the kidney surface	Holds the kidney's shape and protects its surface
Adipose capsule (middle)	A layer of fat around the renal capsule, also called perirenal fat	Cushions the kidney and helps hold it in place
Renal fascia (outermost)	A sheet of dense connective tissue	Anchors the kidney and adrenal gland to the posterior abdominal wall and nearby structures

MEMORY TIP

Go from the kidney outward: capsule, fat, fascia. The capsule sits on the kidney itself, the fat cushions it, and the fascia is farthest out because it ties the whole package to the body wall.

COMPETENCY 134 LECTURE AND LAB

Internal kidney regions

You should be able to: Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney.

When a kidney is cut in half from top to bottom (a frontal section), you see two kinds of tissue and a set of collecting spaces. The tissue that makes urine is called the parenchyma. The collecting spaces carry the finished urine to the ureter.

The **renal cortex** is the outer layer, just under the renal capsule. It looks grainy because it holds the renal corpuscles and the coiled tubules of the nephrons. The **renal medulla** lies deeper. It is made of cone-shaped **renal pyramids**, which look striped because tubules and collecting ducts run side by side through them.

Each pyramid has a wide base facing the cortex and a narrow tip pointing toward the renal sinus. The tip is the **renal papilla**. Urine drains out of tiny openings on the papilla. Between neighboring pyramids, cortex tissue reaches inward as **renal columns**. One pyramid together with the cortex around it makes up one lobe of the kidney.

The cortex and the medulla together form the **parenchyma**, the working tissue of the kidney. All of the nephrons are in the parenchyma.

Each papilla drains into a cup called a **minor calyx**. Several minor calyces join to form a **major calyx**, and the major calyces (usually two or three) join to form the **renal pelvis**. The renal pelvis is a funnel in the renal sinus that narrows into the ureter at the hilum.

WHAT TO FIND ON A SECTIONED KIDNEY

Structure	Where it is	How to recognize it

Structure	Where it is	How to recognize it
Renal cortex	Outer layer, under the capsule	Grainy band around the edge
Renal medulla	Deep to the cortex	Made up of the pyramids
Renal pyramid	In the medulla	Striped cone with its base toward the cortex
Renal papilla	Tip of each pyramid	Points into a minor calyx
Renal column	Between neighboring pyramids	Cortex tissue reaching inward
Parenchyma	Cortex and medulla together	All the tissue that holds nephrons
Minor calyx	Around each papilla	Small cup
Major calyx	Where minor calyces join	Larger branch leading to the pelvis
Renal pelvis	In the renal sinus, at the hilum	Funnel that becomes the ureter
Renal sinus	Space just inside the hilum	Holds the pelvis, calyces, vessels, and fat

MEMORY TIP

Renal columns are cortex. If a strip of tissue sits between two pyramids and looks like the outer layer, it is a renal column, even though it lies deep in the kidney.

COMPETENCY 135 LECTURE AND LAB

Nephron structure

You should be able to: Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron.

The **nephron** is the microscopic unit that makes urine. Each kidney has roughly a million of them. A nephron has two main parts. The **renal corpuscle** filters the blood, and the **renal tubule** turns that filtrate into urine.

The renal corpuscle

The renal corpuscle is the filtering head of the nephron, and it always sits in the cortex. It is made of a ball of capillaries and the cup that surrounds it. The **glomerulus** is the ball of capillaries. Its capillary walls are fenestrated (full of small pores), which makes them very leaky.

The **glomerular capsule** (Bowman's capsule) is the double-walled cup around the glomerulus. Its outer **parietal layer** is simple squamous epithelium and forms the outside wall of the cup. Its inner **visceral layer** hugs the glomerular capillaries and is made of **podocytes**. Podocytes are cells with foot-like extensions that wrap around the capillaries. The narrow gaps between the feet are **filtration slits**.

The **capsular space** lies between the parietal and visceral layers. Filtrate collects in this space and then flows into the tubule.

The renal tubule and collecting duct

The renal tubule leaves the corpuscle as one long tube with three parts. The **proximal convoluted tubule** (PCT) comes first. It is coiled, stays in the cortex, and is lined with cuboidal cells that carry a brush border of microvilli.

Next is the **nephron loop** (loop of Henle), a U-shaped section that dips toward or into the medulla. It has a **descending limb** going down and an **ascending limb** coming back up. The loop returns to the cortex and becomes the **distal convoluted tubule** (DCT), a second coiled section with cuboidal cells and few microvilli.

The DCT empties into a **collecting duct**. Many nephrons drain into the same collecting duct, so it is shared among them. The collecting ducts run down through the pyramids and merge into larger **papillary ducts**, which open on the renal

papilla.

THE PARTS OF A NEPHRON AND ITS COLLECTING DUCT

Structure	Where it is	Key feature
Glomerulus	Renal corpuscle, cortex	Ball of fenestrated capillaries where blood is filtered
Glomerular capsule	Renal corpuscle, cortex	Double-walled cup with a parietal layer and a visceral layer
Podocytes	Visceral layer of the capsule	Foot-like extensions wrap the capillaries and leave filtration slits
Capsular space	Between the two capsule layers	Collects the filtrate
Proximal convoluted tubule	Cortex	First coiled section; cuboidal cells with a brush border
Nephron loop	Dips into the medulla	U-shaped, with a descending limb and an ascending limb
Distal convoluted tubule	Cortex	Second coiled section; few microvilli
Collecting duct	Cortex down through the pyramids	Shared by many nephrons; leads to papillary ducts

Filtrate moves through these parts in one fixed order.

1. **Glomerulus** Blood is filtered out of the capillaries.
2. **Capsular space** The filtrate collects inside the glomerular capsule.
3. **Proximal convoluted tubule** The first coiled section of the tubule.
4. **Descending limb of the nephron loop** The filtrate moves down toward the medulla.
5. **Ascending limb of the nephron loop** The filtrate moves back up toward the cortex.
6. **Distal convoluted tubule** The second coiled section of the tubule.
7. **Collecting duct** Fluid from many nephrons joins here.
8. **Papillary duct** Opens on the renal papilla and drains into a minor calyx.

MEMORY TIP

The segments are named in the order the filtrate meets them: proximal means near the corpuscle, distal means farther away. Learn them in that order and you also know the path of the filtrate.

COMPETENCY 136 LECTURE AND LAB

Cortical vs juxtamedullary nephrons

You should be able to: Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds.

Every nephron has the same parts. Nephrons differ in where the renal corpuscle sits and how deep the nephron loop reaches. The depth of the loop decides which capillary bed runs alongside the tubule.

Cortical nephrons make up about 85 percent of all nephrons. Their renal corpuscles sit in the outer part of the cortex. Their loops are short and barely dip into the outer medulla. They are wrapped by the **peritubular capillaries**, a mesh of capillaries that surrounds the tubules in the cortex.

Juxtamedullary nephrons make up about 15 percent. Juxta means next to, and their renal corpuscles sit in the deepest part of the cortex, right next to the medulla. Their loops are long and reach deep into the medulla, close to the renal

papilla. Long, straight capillaries called the **vasa recta** run down and back up beside these loops. Recta means straight.

Both capillary beds are supplied by efferent arterioles leaving the glomeruli. The long loops of the juxtamedullary nephrons, together with the vasa recta, are what allow the kidney to make concentrated urine.

CORTICAL AND JUXTAMEDULLARY NEPHRONS COMPARED

Feature	Cortical nephrons	Juxtamedullary nephrons
Share of all nephrons	About 85 percent	About 15 percent
Where the renal corpuscle sits	Outer cortex	Deep cortex, next to the medulla
Nephron loop	Short; barely enters the medulla	Long; reaches deep into the medulla
Capillary bed along the tubule	Peritubular capillaries	Vasa recta

MEMORY TIP

Long loop, long straight vessels: the juxtamedullary nephron reaches deep, so its capillaries (the vasa recta) have to reach deep with it.

COMPETENCY 137 LECTURE AND LAB

JGA and filtration membrane

You should be able to: Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back.

The juxtaglomerular apparatus

After the nephron loop climbs back to the cortex, the tubule passes right beside its own renal corpuscle, between the afferent and efferent arterioles. The spot where the tubule and the arterioles touch is the **juxtaglomerular apparatus** (JGA). It checks the fluid in the tubule and the blood pressure in the arteriole, and it helps adjust filtration.

The **macula densa** is a patch of tall, closely packed cells in the tubule wall, where the ascending limb of the loop ends and the distal convoluted tubule begins. Macula densa means dense spot, which describes how the crowded cells look on a slide. These cells sense the amount of sodium chloride in the fluid passing by.

The **juxtaglomerular cells** (JG cells, also called **granular cells**) are modified smooth muscle cells, mostly in the wall of the afferent arteriole. Their granules hold the enzyme renin, which they release when blood pressure drops.

THE JUXTAGLOMERULAR APPARATUS

Structure	Where it is	What it does
Juxtaglomerular apparatus	Where the tubule touches the afferent and efferent arterioles of its own corpuscle	Monitors the tubule fluid and the arteriole and helps adjust filtration
Macula densa	Tubule wall, where the ascending limb meets the distal convoluted tubule	Senses the sodium chloride level of the fluid
Juxtaglomerular (granular) cells	Wall of the afferent arteriole	Release renin when blood pressure is low

The filtration membrane

Blood in the glomerulus is filtered into the capsular space across the **filtration membrane**. Fluid crosses three layers in a set order, and each layer holds back a different size of particle.

THE THREE LAYERS OF THE FILTRATION MEMBRANE, FROM THE BLOOD SIDE TO THE CAPSULAR SPACE

Layer	What it is	What it holds back
Fenestrated endothelium	The pored wall of the glomerular capillary	Blood cells and platelets
Basement membrane	The layer between the capillary wall and the podocytes	Large plasma proteins
Filtration slits of the podocytes	The gaps between podocyte feet, bridged by thin slit membranes	Most of the smaller proteins that get past the basement membrane

What passes all three layers is the filtrate: water and small dissolved substances such as ions, glucose, and wastes like urea. Healthy filtrate has almost no protein and no blood cells. Protein or blood in the urine is a sign that the filtration membrane is damaged.

MEMORY TIP

Each layer catches something smaller than the one before it: cells first, then big proteins, then the smaller proteins.

COMPETENCY 138 LECTURE AND LAB

Kidney blood supply pathway

You should be able to: Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order.

The kidneys receive about a quarter of the blood the heart pumps at rest. Blood follows a fixed chain of arteries inward, passes through two capillary beds, and leaves through a chain of veins that runs back along the same route.

The names of the vessels tell you where they run. A pyramid with its surrounding cortex is a lobe, so **interlobar** vessels run between lobes, up through the renal columns. **Arcuate** means arched, and these vessels arch over the bases of the pyramids at the border between the cortex and medulla. **Cortical radiate** vessels radiate outward into the cortex. Some books call them interlobular vessels.

1. **Renal artery** Branches off the abdominal aorta and enters the kidney at the hilum.
2. **Segmental arteries** Branches of the renal artery inside the renal sinus.
3. **Interlobar arteries** Run up through the renal columns, between the pyramids.
4. **Arcuate arteries** Arch over the bases of the pyramids, along the border between cortex and medulla.
5. **Cortical radiate arteries** Also called interlobular arteries; radiate outward into the cortex.
6. **Afferent arteriole** Carries blood into one glomerulus.
7. **Glomerulus** The first capillary bed, where blood is filtered.
8. **Efferent arteriole** Carries blood out of the glomerulus.
9. **Peritubular capillaries and vasa recta** The second capillary bed. Peritubular capillaries wrap the tubules in the cortex, and the vasa recta run beside the long loops in the medulla.
10. **Cortical radiate veins** Drain the cortex and run back toward the medulla.
11. **Arcuate veins** Arch over the bases of the pyramids.
12. **Interlobar veins** Run down through the renal columns toward the renal sinus.
13. **Renal vein** Leaves the kidney at the hilum and empties into the inferior vena cava.

This path is unusual in one way. In most of the body, a capillary bed drains into a venule. The glomerulus has an arteriole on both sides, the afferent arteriole going in and the efferent arteriole coming out. Blood then passes through a second capillary bed before it reaches any vein.

MEMORY TIP

Blood meets two capillary beds in the kidney, one after the other: the glomerulus, where filtration happens, and then the peritubular capillaries and vasa recta around the tubules. The efferent arteriole is the vessel between them. On the way out, the veins repeat the artery names in reverse: cortical radiate, arcuate, interlobar, renal.

COMPETENCY 139 LECTURE AND LAB

Ureters, bladder, urethra and urine path

You should be able to: Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra.

Once fluid leaves the collecting duct it is urine, and nothing along the rest of the path changes it much. The rest of the system collects the urine, carries it, stores it, and releases it.

The path of urine

1. **Collecting duct** Gathers fluid from many nephrons and carries it through the pyramid.
2. **Papillary duct** Opens on the renal papilla.
3. **Minor calyx** Cup that catches urine from one papilla.
4. **Major calyx** Forms where several minor calyces join.
5. **Renal pelvis** Funnel that collects urine from the major calyces.
6. **Ureter** Carries urine down to the bladder.
7. **Urinary bladder** Stores the urine.
8. **Urethra** Carries urine out of the body.
9. **External urethral orifice** The opening to the outside.

The ureters

Each **ureter** is a muscular tube about 30 cm long. It leaves the renal pelvis at the hilum, runs down the posterior abdominal wall behind the peritoneum, and enters the posterior wall of the bladder. The ureter passes through the bladder wall at a slant. When the bladder fills, pressure squeezes this slanted part closed, which keeps urine from flowing back up toward the kidney.

The ureter wall has three layers. The inner mucosa is lined with **transitional epithelium** (urothelium), which can stretch. The middle layer is smooth muscle, which pushes urine along by peristalsis. The outer layer is a fibrous **adventitia**.

The urinary bladder

The **urinary bladder** is a hollow, muscular sac in the pelvic cavity, just behind the pubic symphysis. In males it sits in front of the rectum and above the prostate. In females it sits in front of the vagina and below the uterus. Only its upper surface is covered by peritoneum.

THE LAYERS OF THE BLADDER WALL, FROM THE INSIDE OUT

Layer	What it is	Key feature
Mucosa	Transitional epithelium (urothelium) on a layer of connective tissue (lamina propria)	Stretches as the bladder fills; folds into **rugae** when the bladder is empty
Detrusor muscle	The muscular layer, made of layers of smooth muscle	Contracts to squeeze urine out
Adventitia	Outer fibrous connective tissue	Over the upper surface, the peritoneum (a serosa) covers the bladder

The **trigone** is a smooth triangle on the floor of the bladder. Its three corners are the two **ureteral openings** at the back and the **internal urethral orifice** at the front, where the urethra begins. The trigone has no rugae, so it stays smooth even when the bladder is empty.

Two sphincters control the outflow. The **internal urethral sphincter** is a thickening of smooth muscle at the neck of the bladder, where the urethra begins. It is involuntary. The **external urethral sphincter** is skeletal muscle that surrounds the urethra as it passes through the pelvic floor. It is voluntary and gives conscious control over urination.

The urethra

The **urethra** carries urine from the bladder to the outside. It is very different in females and males.

THE FEMALE AND MALE URETHRA COMPARED

Feature	Female urethra	Male urethra
Length	About 4 cm	About 20 cm
What it carries	Urine only	Urine and semen
Parts	One short tube	**Prostatic urethra** through the prostate, **membranous urethra** through the deep muscles of the perineum, and **spongy urethra** through the corpus spongiosum of the penis
External urethral orifice	Between the clitoris and the vaginal opening	At the tip of the glans penis

COMPETENCY 140 LECTURE

Clinical anatomy of urinary disorders

You should be able to: Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects.

Urinary disorders make the most sense when you place each one on the structures you already know. Some damage the filtering parts of the nephron. Others block, infect, or weaken the tubes that carry and hold urine.

Any blockage along the path makes urine back up into the parts above it. When urine backs up into the renal pelvis and calyces, they swell. This swelling is called **hydronephrosis**.

URINARY DISORDERS AND THE STRUCTURES THEY AFFECT

Disorder	Structure or pathway affected	What changes

Disorder	Structure or pathway affected	What changes
Nephroptosis	Adipose capsule and renal fascia; the ureter below	The kidney drops out of its normal position, often when its cushion of fat is thin or lost. The ureter can kink and block urine flow.
Kidney stones (renal calculi)	Form in the calyces and renal pelvis; lodge in the ureter	Minerals harden into stones. A stone tends to stick at the narrow points of the ureter: where it leaves the renal pelvis, where it crosses the iliac vessels at the pelvic brim, and where it enters the bladder.
Urinary tract infection	Urethra, bladder, and, if it climbs, the ureter and kidney	Bacteria, most often from the intestine (such as E. coli), climb up the urine path and inflame its lining. Infection of the bladder is cystitis. Infection that reaches the renal pelvis and kidney tissue is pyelonephritis.
Glomerulonephritis	Glomeruli	The glomeruli become inflamed, which damages the filtration membrane. Protein and blood can leak into the urine.
Renal failure	Nephrons of both kidneys	The kidneys lose their ability to filter the blood, either suddenly (acute) or slowly over time (chronic).
Polycystic kidney disease	Renal parenchyma (the tubules)	An inherited disorder in which many fluid-filled cysts form and enlarge the kidneys, crowding out working tissue.
Urinary incontinence	Detrusor muscle, urethral sphincters, and pelvic floor	Loss of voluntary control over urination. In stress incontinence a weak pelvic floor and external sphincter leak with coughing or lifting. In urge incontinence the detrusor contracts too readily. In overflow incontinence the bladder cannot empty fully, so it overfills and leaks. In functional incontinence the urinary organs work, but the person cannot reach a toilet in time.

POSITION AND PATHWAY PREDICT THE PROBLEM

The kidney depends on its **adipose capsule** to stay in place. When that fat is lost, the kidney can drop, and the ureter can kink and block the flow of urine.

Urine travels one narrow path. A **kidney stone** often causes little pain while it sits in the renal pelvis. Pain usually starts when the stone moves into the **ureter**, where it scrapes and blocks a tube only a few millimeters wide. Where the stone sits along that path shapes where the pain is felt.

The two urethras differ, and so does the risk. The **female urethra** is short and opens close to the anus, which gives bacteria a shorter climb to the bladder. This is one reason urinary tract infections are more common in women.

CHAPTER 2

The Male Reproductive System

REPRODUCTIVE

COMPETENCIES 141 TO 145

Covers the scrotum and testes, the duct system and the path of sperm, the structure of a sperm cell, the accessory glands, and the penis. The focus is on the structures and the job each one does.

COMPETENCIES

1. **141** Scrotum and testis structure
2. **142** Sperm duct system and pathway
3. **143** Sperm cell structure
4. **144** Accessory glands and penis
5. **145** Clinical anatomy of male disorders

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The male reproductive system

drsrennie-stack.github.io/new-build-bio4-solano/reproductive-male-concept-videos.html

COMPETENCY 141 LECTURE AND LAB

Scrotum and testis structure

You should be able to: Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules, sustentacular and interstitial cells, rete testis, and efferent ductules.

The male reproductive system makes sperm and delivers them. Its parts sort into four groups: the gonads, a duct system, the accessory glands, and the supporting structures. This section covers the gonads, the **testes**, and the pouch that holds them.

THE FOUR GROUPS OF PARTS

Group	What it includes
Testes	The male gonads; they produce sperm and secrete hormones
Duct system	The epididymis, ductus deferens, ejaculatory ducts, and urethra, which store, mature, and carry sperm
Accessory glands	The seminal vesicles, prostate, and bulbourethral glands, which add fluid to form semen
Supporting structures	The scrotum and the penis

The scrotum and its two muscles

The **scrotum** is a pouch of skin and subcutaneous tissue that hangs behind the penis, outside the pelvic cavity. A ridge down its middle, the **raphe**, marks where the **scrotal septum** divides the inside into two compartments, one for each testis. The testes sit outside the body because sperm production needs a temperature a few degrees below the body core.

Two muscles adjust that temperature, and they are different tissues. The **dartos muscle** is a thin layer of smooth muscle in the subcutaneous tissue of the scrotum. When it contracts, the scrotal skin wrinkles and tightens, which cuts heat loss. The **cremaster muscle** is skeletal muscle. It comes off the internal oblique muscle of the abdominal wall, travels down in the spermatic cord, and loops around each testis. When it contracts, it pulls the testis up toward the warmer body wall. In warm conditions both muscles relax and the testes hang lower.

The coverings of the testis

Each **testis** is an oval organ wrapped in two layers called tunics. The outer one, the **tunica vaginalis**, is a serous membrane with a parietal layer and a visceral layer. It comes from a pouch of peritoneum that the testis brought with it when it moved down into the scrotum before birth, and it covers the front and sides of the testis. Under it is the **tunica albuginea**, a tough white capsule of dense connective tissue. The tunica albuginea sends thin walls (septa) inward that divide the testis into a few hundred wedge-shaped **lobules**.

Inside the testis

Each lobule holds tightly coiled **seminiferous tubules**. These tubules are the only place sperm are made. On a slide, a seminiferous tubule is a round or oval profile with a thick wall of cells around a small central lumen. The developing sperm cells are stacked in that wall, the least mature at the outer edge and the most mature next to the lumen.

Sustentacular cells (Sertoli cells) are tall cells that reach from the outer edge of the tubule to its lumen. They surround, support, and nourish the developing sperm cells. Tight junctions between neighboring sustentacular cells form the **blood–testis barrier**, which keeps substances in the blood away from the developing sperm. **Interstitial cells** (Leydig cells) sit in the connective tissue between the tubules, outside the tubule walls. They produce testosterone.

The tubules empty toward the back of the testis. Short **straight tubules** carry sperm into the **rete testis**, a network of small channels in the connective tissue at the posterior side of the testis. From the rete testis, a set of tiny **efferent ductules** leave the testis and carry the sperm into the head of the epididymis.

THE SCROTUM AND THE TESTIS, FROM THE OUTSIDE IN

Structure	Where it is	What it does
Scrotum	Skin pouch behind the penis, outside the pelvic cavity	Holds the testes; the scrotal septum gives each testis its own compartment
Dartos muscle	Smooth muscle in the subcutaneous tissue of the scrotum	Wrinkles the scrotal skin to reduce heat loss
Cremaster muscle	Skeletal muscle in the spermatic cord, looping around the testis	Raises the testis toward the body
Tunica vaginalis	Serous membrane over the front and sides of the testis	Lets the testis move smoothly inside the scrotum
Tunica albuginea	Dense white capsule directly on the testis	Encloses the testis and divides it into lobules
Seminiferous tubules	Coiled tubules inside each lobule	Produce sperm
Sustentacular cells	Inside the wall of each seminiferous tubule	Support developing sperm; form the blood-testis barrier
Interstitial cells	In the connective tissue between the tubules	Produce testosterone
Rete testis	Network of channels at the posterior side of the testis	Collects sperm from the tubules
Efferent ductules	Small ducts between the rete testis and the epididymis	Carry sperm out of the testis into the epididymis

MEMORY TIP

The names tell you where to look on the slide. Sustentacular cells sustain the sperm, so they sit inside the tubule wall with the sperm. Interstitial cells sit in the interstitial spaces, between the tubules.

COMPETENCY 142 LECTURE AND LAB

Sperm duct system and pathway

You should be able to: Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents.

Only the seminiferous tubules make sperm. Every duct after them stores, matures, or carries the sperm. The route runs in one direction: up out of the scrotum, into the pelvis behind the bladder, and then down through the prostate and the penis to the outside.

Epididymis

The **epididymis** is a comma-shaped structure on the posterior side of each testis. Its head sits at the top, where the efferent ductules enter. Its body runs down the back of the testis, and its tail sits at the bottom. Inside is a single duct about 6 meters long, coiled tightly enough to fit in a few centimeters. Sperm are stored here, and they gain the ability to swim as they move through it.

Ductus deferens

The **ductus deferens** (vas deferens) begins at the tail of the epididymis. It is a long duct with a thick wall of smooth muscle, so it feels like a firm cord through the scrotal skin. It climbs in the spermatic cord, passes through the inguinal canal into the pelvic cavity, crosses over the ureter, and runs down the posterior surface of the bladder. Its last part widens into the **ampulla of the ductus deferens**.

Ejaculatory duct and urethra

Behind the bladder, each ampulla joins the duct of a seminal vesicle. The two together form the short **ejaculatory duct**. There is one on each side. Each ejaculatory duct passes through the prostate and empties into the urethra.

The male **urethra** carries both urine and semen. It has three parts. The **prostatic urethra** runs through the prostate, just below the bladder. The **membranous urethra** is the short part that passes through the muscles of the pelvic floor (the urogenital diaphragm). The **spongy urethra** (penile urethra) is the longest part. It runs through the corpus spongiosum of the penis and opens at the **external urethral orifice** on the glans.

Spermatic cord

The **spermatic cord** is the bundle that suspends each testis and carries its duct, vessels, and nerves between the scrotum and the abdomen. It passes through the **inguinal canal**, a slanted passage through the lower anterior abdominal wall. The cord holds these structures:

- The ductus deferens
- The **testicular artery**, a branch of the abdominal aorta
- The **pampiniform plexus**, a network of veins wrapped around the testicular artery; it cools the blood flowing in to the testis, and its veins join to form the **testicular vein**
- Lymphatic vessels and nerves
- The cremaster muscle and layers of fascia that wrap the whole cord

THE DUCTS, IN ORDER

Structure	Where it is	What it does
Epididymis	Posterior side of the testis	Stores sperm; sperm gain the ability to swim
Ductus deferens	From the epididymis, up the spermatic cord, through the inguinal canal, behind the bladder	Carries sperm into the pelvis
Ejaculatory duct	Inside the prostate	Joins the ductus deferens and seminal vesicle duct; empties into the prostatic urethra
Prostatic urethra	Inside the prostate	First part of the urethra

Structure	Where it is	What it does
Membranous urethra	Through the pelvic floor muscles	Short middle part of the urethra
Spongy urethra	Inside the corpus spongiosum of the penis	Longest part; opens at the external urethral orifice
Spermatic cord	From the scrotum through the inguinal canal	Carries the ductus deferens, vessels, and nerves of the testis

Sperm follow this route from the testis to the outside.

1. **Seminiferous tubules** sperm are made in the walls of the tubules inside the testis
2. **Rete testis** straight tubules carry the sperm into this network at the back of the testis
3. **Efferent ductules** small ducts carry the sperm out of the testis
4. **Epididymis** sperm enter the head, are stored, and mature as they move to the tail
5. **Ductus deferens** sperm travel up the spermatic cord, through the inguinal canal, over the ureter, and behind the bladder to the ampulla
6. **Ejaculatory duct** the ductus deferens joins the duct of the seminal vesicle and passes through the prostate
7. **Prostatic urethra** sperm enter the urethra inside the prostate
8. **Membranous urethra** sperm pass through the pelvic floor
9. **Spongy urethra** sperm pass through the corpus spongiosum of the penis
10. **External urethral orifice** semen leaves the body at the tip of the glans

MEMORY TIP

SEVEN UP gives the main stops in order: Seminiferous tubules, Epididymis, Vas deferens, Ejaculatory duct, (N is for nothing), Urethra, Penis.

COMPETENCY 143 LECTURE AND LAB

Sperm cell structure

You should be able to: Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell.

A mature **sperm cell** (spermatozoon) is built for one job: reaching an oocyte and delivering its chromosomes. It has two main parts, a head and a tail. The tail is a long **flagellum**. From front to back, the tail has four parts: the neck, the middle piece, the principal piece, and the end piece.

The **head** is a flattened oval. Most of it is the **nucleus**, which holds 23 chromosomes, half the number in a body cell. There is almost no cytoplasm. The **acrosome** is a cap that covers the front of the nucleus. It is a sac of enzymes that lets the sperm digest a path through the layers around the oocyte.

The **neck** is the short, narrow part just behind the head. It holds the **centrioles**, and the core of microtubules that runs the length of the tail starts here.

The **middle piece** is packed with **mitochondria** wound in a spiral around the core of the tail. They make the ATP that powers the tail. The **principal piece** is the longest part of the tail and does most of the whipping that moves the cell forward. The **end piece** is the short, thin tip at the very end of the tail.

On a slide of the testis, you see sperm heads and tails crowded in the lumen of the seminiferous tubules. On a sperm smear, the head is the dark, densely stained oval, and the tail is the long thin line behind it.

THE PARTS OF A SPERM CELL, FRONT TO BACK

Part	Where it is	What it holds or does
Head	Front of the cell	Holds the nucleus with 23 chromosomes
Acrosome	Cap over the front of the head	Holds enzymes that help the sperm get through the layers around the oocyte
Neck	Just behind the head	Holds the centrioles; the tail core starts here
Middle piece	First long stretch of the tail	Packed with mitochondria that power movement
Principal piece	Most of the tail	Longest part; drives the swimming
End piece	Tip of the tail	Short, thin final part

MEMORY TIP

Name the parts front to back: head with its acrosome cap, then neck, middle piece, principal piece, end piece. The middle piece is the only part packed with mitochondria.

COMPETENCY 144 LECTURE AND LAB

Accessory glands and penis

You should be able to: Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis.

The accessory glands

Sperm are only a small part of semen. Three sets of accessory glands add most of the fluid. The mix of sperm and these secretions is **semen**. Where each gland sits tells you where its fluid joins the duct system.

The **seminal vesicles** are a pair of glands on the posterior surface of the bladder, beside the ampullae of the ductus deferens. The duct of each seminal vesicle joins the ductus deferens to form the ejaculatory duct. Their fluid is alkaline and rich in fructose, a sugar the sperm use for energy. It makes up about 60 percent of the volume of semen.

The **prostate** is a single gland about the size of a walnut. It sits just below the bladder and wraps around the prostatic urethra. Its back surface lies against the rectum. Many small ducts carry its milky fluid into the prostatic urethra, where it helps the sperm stay active.

The **bulbourethral glands** (Cowper glands) are a pair of pea-sized glands just below the prostate, beside the membranous urethra in the pelvic floor. Their ducts open into the spongy urethra. They release a clear mucus before ejaculation that neutralizes traces of acidic urine in the urethra and lubricates the tip of the penis.

THE ACCESSORY GLANDS

Gland	Number	Where it is	Empties into	What it adds
Seminal vesicles	Paired	Posterior surface of the bladder	Ejaculatory duct	Alkaline, fructose-rich fluid; about 60 percent of semen
Prostate	Single	Just below the bladder, around the prostatic urethra	Prostatic urethra	Milky fluid that helps sperm stay active
Bulbourethral glands	Paired	Below the prostate, beside the membranous urethra	Spongy urethra	Clear alkaline mucus that neutralizes urine traces and lubricates

The penis

The penis carries the spongy urethra and delivers semen. It has three regions. The **root** is the attached part, fixed to the pelvis in the perineum. It is made of the **bulb of the penis**, which is the back end of the corpus spongiosum, and the two **crura**, which are the back ends of the corpora cavernosa and attach to the ischial and pubic rami. The **body** (shaft) is the free, movable part. The **glans penis** is the enlarged tip, formed by the expanded end of the corpus spongiosum. The external urethral orifice opens on the glans. The **prepuce** (foreskin) is the loose fold of skin that covers the glans. It is the part removed in circumcision.

Three columns of **erectile tissue** run the length of the body. Erectile tissue is a sponge of blood spaces inside connective tissue and smooth muscle. When these spaces fill with blood, the penis becomes erect. The two **corpora cavernosa** lie side by side on the dorsal side. The single **corpus spongiosum** lies on the ventral side and surrounds the spongy urethra.

THE REGIONS OF THE PENIS AND ITS ERECTILE COLUMNS

Structure	What it is
Root	The attached part; the bulb of the penis and the two crura
Body	The free, movable shaft
Glans penis	The enlarged tip; the expanded end of the corpus spongiosum
Prepuce	The foreskin; the loose fold of skin over the glans
Corpora cavernosa	The two dorsal columns of erectile tissue; their back ends form the crura
Corpus spongiosum	The single ventral column; surrounds the spongy urethra and forms the bulb and the glans

MEMORY TIP

In a cross section of the penis, find the urethra first. The column wrapped around it is the corpus spongiosum. The two larger columns above it are the corpora cavernosa.

COMPETENCY 145 LECTURE

Clinical anatomy of male disorders

You should be able to: Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position.

Most male reproductive disorders make sense once you know where the structure sits. The testes form inside the abdomen and move down into the scrotum. The prostate wraps around the urethra. The veins of the testis run up a long cord. Each of these positions explains a disorder.

Before birth, the testes develop high on the posterior abdominal wall, near the kidneys. That is why each testicular artery branches from the abdominal aorta. Late in fetal development, each testis moves down and passes through the inguinal canal into the scrotum, pulling its duct and vessels behind it. In **cryptorchidism**, one or both testes stop along this route, in the abdomen or in the inguinal canal. An undescended testis stays at core body temperature, which impairs sperm production and raises the risk of testicular cancer.

The prostate sits just below the bladder and surrounds the prostatic urethra. In **benign prostatic hyperplasia** (BPH), the prostate enlarges without cancer, most often in the tissue right around the urethra. The enlarged gland squeezes the prostatic urethra, so the urine stream weakens and the bladder may not empty fully. BPH is common in older men. **Prostate cancer** most often starts in the outer, posterior part of the gland. Because the back of the prostate lies against the rectum, a clinician can feel it through the rectal wall.

The inguinal canal is a weak spot in the abdominal wall, left by the descent of the testis. In an **inguinal hernia**, a loop of intestine pushes into the canal and can follow the spermatic cord down into the scrotum. In a **hydrocele**, serous fluid

collects between the two layers of the tunica vaginalis. In a **varicocele**, the veins of the pampiniform plexus swell and twist. Varicoceles are more common on the left, where the testicular vein drains into the left renal vein.

MALE DISORDERS BY STRUCTURE AND POSITION

Disorder	Structure affected	Where it is	What changes
Cryptorchidism	Testis	Abdomen or inguinal canal, instead of the scrotum	The testis fails to descend and stays at core body temperature
Benign prostatic hyperplasia	Prostate	Below the bladder, around the prostatic urethra	Non-cancerous enlargement that squeezes the urethra
Prostate cancer	Prostate	Usually the posterior part, against the rectum	Malignant tumor of the gland
Testicular cancer	Testis	In the scrotum	Malignant tumor, usually from the sperm-forming cells of the seminiferous tubules
Hydrocele	Tunica vaginalis	Around the testis	Serous fluid collects between its two layers and the scrotum swells
Varicocele	Pampiniform plexus	Spermatic cord, above the testis	The veins swell and twist, more often on the left
Inguinal hernia	Inguinal canal	Lower anterior abdominal wall	A loop of intestine pushes into the canal along the spermatic cord

WHY THE TESTES HANG OUTSIDE

Sperm production needs a temperature a few degrees below the body core. That is why the testes sit in the **scrotum** and why the **cremaster** and **dartos** muscles adjust their position. In **cryptorchidism**, an undescended testis stays at body temperature, and if it is not corrected the warmth can damage the sperm-forming cells.

The duct system shares its final stretch with the urinary tract. Because the **urethra** passes straight through the **prostate**, an enlarged prostate squeezes the urethra and makes urination difficult. This is how a reproductive organ produces a urinary symptom.

CHAPTER 3

The Female Reproductive System

REPRODUCTIVE

COMPETENCIES 146 TO 150

Covers the ovaries, the uterine tubes and the path of the ovum, the uterus, the vagina and external genitalia, and the mammary glands. The focus is on the structures and the job each one does.

COMPETENCIES

1. **146** Ovary structure and follicle stages
2. **147** Uterine tubes and ovum pathway
3. **148** Uterus regions, ligaments, and wall
4. **149** Vagina and external genitalia
5. **150** Mammary gland structure

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The female reproductive system

drsrennie-stack.github.io/new-build-bio4-solano/reproductive-female-concept-videos.html

COMPETENCY 146 LECTURE AND LAB

Ovary structure and follicle stages

You should be able to: Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans.

The female reproductive system makes oocytes, receives sperm, and supports a developing fetus. Its organs are the ovaries, the uterine tubes, the uterus, the vagina, the external genitalia, and the mammary glands.

THE ORGANS OF THE SYSTEM

Structure	What it is
Ovaries	The female gonads; they produce oocytes and secrete hormones
Uterine tubes	The fallopian tubes; they carry the oocyte toward the uterus
Uterus	The muscular organ where an embryo implants and a fetus develops
Vagina	The canal from the cervix to the outside; the birth canal
Vulva	The external genitalia
Mammary glands	The milk-producing glands of the breasts

Where the ovaries sit and what holds them

The **ovaries** are the paired female gonads. Each is an almond-shaped organ against the lateral wall of the pelvic cavity, one on each side of the uterus, just below the open end of a uterine tube. Three ligaments hold each ovary in place.

The **broad ligament** is a large fold of peritoneum that drapes over the uterus, the uterine tubes, and the ovaries. A short fold of it, the **mesovarium**, attaches the ovary to the back of the broad ligament. The **ovarian ligament** is a cord of fibrous tissue that ties the ovary to the side of the uterus, just below where the uterine tube joins it. The **suspensory ligament** runs from the ovary up to the lateral pelvic wall. It carries the **ovarian artery** and **ovarian vein** to and from the ovary. Like the testicular artery, the ovarian artery branches from the abdominal aorta.

THE LIGAMENTS OF THE OVARY

Ligament	What it connects
Broad ligament (mesovarium)	Ovary to the back of the broad ligament, the fold of peritoneum over the uterus and tubes
Ovarian ligament	Ovary to the side of the uterus
Suspensory ligament	Ovary to the lateral pelvic wall; carries the ovarian artery and vein

The layers of the ovary

Read the ovary from the outside in. The **germinal epithelium** (ovarian surface epithelium) is a single layer of cuboidal cells covering the ovary. Despite its name, it does not make oocytes. Under it is the **tunica albuginea**, a capsule of dense connective tissue, with the same name as the capsule of the testis. Next is the **ovarian cortex**, which holds the follicles. The center is the **ovarian medulla**, loose connective tissue with the blood vessels, lymphatic vessels, and nerves.

The follicle stages

An **ovarian follicle** is an oocyte plus the cells around it. The cells next to the oocyte are follicle cells, called **granulosa cells** once they become cuboidal. In later stages, connective tissue cells outside them form a layer called the **theca**. Follicles grow through a set series of stages, so on a slide you name each one by its size, its layers of cells, and whether it has a fluid space.

A **primordial follicle** is a primary oocyte with one layer of flat follicle cells. These are the smallest and most numerous follicles, in the outer cortex. In a **primary follicle**, the follicle cells become cuboidal granulosa cells and a clear layer, the **zona pellucida**, forms around the oocyte. A **secondary follicle** has several layers of granulosa cells, a theca around the outside, and small pockets of fluid between the granulosa cells. A **mature follicle** (vesicular or Graafian follicle) has one large fluid-filled cavity, the **antrum**. Its oocyte sits on a mound of granulosa cells to one side, ringed by the **corona radiata**. It is big enough to bulge from the surface of the ovary. Just before ovulation, its oocyte becomes a secondary oocyte.

At **ovulation**, the mature follicle ruptures and releases the secondary oocyte with its corona radiata. The granulosa and theca cells left behind become the **corpus luteum** (yellow body), a large glandular body that secretes hormones. If no pregnancy occurs, the corpus luteum breaks down into the **corpus albicans** (white body), a pale scar of fibrous tissue. Fluid-filled **ovarian cysts** often form from a follicle or a corpus luteum that does not go away.

FOLLICLE STAGES IN ORDER

Stage	Oocyte	Cells around it	How to spot it
Primordial follicle	Primary oocyte	One layer of flat follicle cells	Smallest; many, in the outer cortex
Primary follicle	Primary oocyte	Cuboidal granulosa cells; zona pellucida forms	Oocyte ringed by cuboidal cells
Secondary follicle	Primary oocyte	Several layers of granulosa cells and a theca	Small fluid pockets between the granulosa cells
Mature follicle	Primary oocyte; it becomes a secondary oocyte just before ovulation	Granulosa cells, corona radiata, theca	One large antrum; bulges from the ovary
Corpus luteum	None (released at ovulation)	Granulosa and theca cells left behind	Large, pale or yellow glandular body

Stage	Oocyte	Cells around it	How to spot it
Corpus albicans	None	Fibrous scar tissue	Pale, white scar in the cortex

MEMORY TIP

The suspensory ligament suspends the ovary from the pelvic wall and carries its vessels. The ovarian ligament ties the ovary to the uterus.

COMPETENCY 147 LECTURE AND LAB

Uterine tubes and ovum pathway

You should be able to: Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus.

The **uterine tubes** (fallopian tubes) are a pair of tubes about 10 cm long. Each runs from an upper corner of the uterus out toward an ovary, inside the upper edge of the broad ligament. This part of the broad ligament is the **mesosalpinx**. The tube carries the oocyte toward the uterus, and it is where fertilization usually happens.

The outer end of the tube is not attached to the ovary. At ovulation the oocyte is released into the pelvic cavity, and the tube has to catch it. The tube has four named regions, from the ovary toward the uterus.

The **infundibulum** is the funnel-shaped outer end, open to the pelvic cavity near the ovary. Its edge is fringed with **fimbriae**, finger-like projections that drape over the ovary. Around ovulation the fimbriae sweep across the surface of the ovary and draw the oocyte into the funnel. The **ampulla** is the widest and longest part of the tube. Fertilization usually happens here. The **isthmus** is the narrow, thick-walled part that joins the uterus. A short **intramural part** then passes through the uterine wall and opens into the uterine cavity.

The lining of the tube has ciliated cells and secretory cells. The cilia beat toward the uterus, and a layer of smooth muscle in the wall squeezes in waves. Together they move the oocyte, or the early embryo, toward the uterus over a few days.

THE REGIONS OF THE UTERINE TUBE, FROM THE OVARY TO THE UTERUS

Region	What it is
Fimbriae	Finger-like projections on the edge of the infundibulum that sweep the oocyte into the tube
Infundibulum	The funnel-shaped outer end, open to the pelvic cavity near the ovary
Ampulla	The widest and longest part; the usual site of fertilization
Isthmus	The narrow, thick-walled part next to the uterus
Intramural part	The short part that passes through the uterine wall into the uterine cavity

The oocyte follows this route from the ovary to the uterus.

1. **Mature follicle** in the ovarian cortex, the follicle ruptures at ovulation and releases the secondary oocyte into the pelvic cavity
2. **Fimbriae** the fimbriae sweep over the ovary and draw the oocyte in
3. **Infundibulum** the funnel carries the oocyte into the tube
4. **Ampulla** the oocyte moves through the widest part of the tube, where sperm usually meet and fertilize it

5. **Isthmus** the oocyte or early embryo moves through the narrow part of the tube
6. **Intramural part** it passes through the uterine wall
7. **Uterine cavity** it arrives in the cavity of the uterus; an embryo implants in the endometrium, usually in the upper part of the body of the uterus

MEMORY TIP

The tube narrows as it nears the uterus: fimbriae and infundibulum (the open funnel), then the wide ampulla, then the narrow isthmus. An isthmus of land is a narrow neck, and so is this one.

AN OPEN PATHWAY TO THE PELVIC CAVITY

The female tract is open to the pelvic cavity. The **infundibulum** opens near the ovary and does not seal onto it. That opening is how the fimbriae catch an oocyte. It is also how an infection can climb from the vagina through the cervix, uterus, and uterine tubes into the pelvic cavity. This spreading infection is **pelvic inflammatory disease** (PID).

In an **ectopic pregnancy**, an embryo implants outside the uterine cavity. Most implant in the uterine tube, usually in the ampulla. This happens when the embryo is held up inside the tube, for example where scarring from earlier PID narrows it. The thin wall of the tube cannot stretch to hold a growing embryo. In rare cases an embryo implants in the pelvic or abdominal cavity, which is possible because the infundibulum is open.

COMPETENCY 148 LECTURE AND LAB

Uterus regions, ligaments, and wall

You should be able to: Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium.

The **uterus** is a hollow, thick-walled, pear-shaped organ in the pelvic cavity. It sits behind the urinary bladder and in front of the rectum. In most women it tips forward over the bladder. It receives the embryo, holds the developing fetus, and contracts during childbirth.

Regions

The **fundus** is the rounded top, above the points where the uterine tubes enter. The **body** is the main central part. The space inside it is the **uterine cavity**. Below the body, a short narrowed region called the isthmus of the uterus leads to the **cervix**, the narrow lower part that projects into the vagina. The inside of the cervix is the **cervical canal**. It opens into the uterine cavity at the **internal os** and into the vagina at the **external os**.

THE REGIONS OF THE UTERUS

Region	What it is
Fundus	The rounded top, above the entry of the uterine tubes
Body	The main central part; holds the uterine cavity
Cervix	The narrow lower part that projects into the vagina
Uterine cavity	The space inside the body of the uterus
Cervical canal	The channel through the cervix, from the internal os to the external os

Supporting ligaments

Four paired ligaments hold the uterus, with help from the muscles of the pelvic floor. The **broad ligaments** are a double fold of peritoneum from the sides of the uterus to the lateral pelvic walls. The part of it beside the uterus is the mesometrium; the same sheet also holds the uterine tubes (mesosalpinx) and the ovaries (mesovarium). The **round ligaments** are fibrous cords that leave the uterus just below each uterine tube, run forward through the inguinal canal, and end in the labia majora. They help keep the uterus tipped forward. The **cardinal ligaments** (transverse cervical ligaments) run from the cervix and upper vagina to the lateral pelvic walls. They are the main support that keeps the uterus from sliding down. The **uterosacral ligaments** run from the cervix back to the sacrum.

THE LIGAMENTS OF THE UTERUS

Ligament	Runs from	Runs to
Broad ligaments	Sides of the uterus	Lateral pelvic walls and floor
Round ligaments	Uterus, just below each uterine tube	Through the inguinal canal to the labia majora
Cardinal ligaments	Cervix and upper vagina	Lateral pelvic walls
Uterosacral ligaments	Cervix	Sacrum

MEMORY TIP

The round ligament takes the same route through the inguinal canal that the spermatic cord takes in a male.

The three layers of the wall

The **perimetrium** is the outer serous layer. It is part of the visceral peritoneum. The **myometrium** is the thick middle layer of smooth muscle, thickest in the fundus. It contracts during childbirth. The **endometrium** is the inner mucous lining, where an embryo implants. It has two layers. The **stratum functionalis** (functional layer), next to the cavity, is shed during menstruation. The **stratum basalis** (basal layer), next to the myometrium, stays in place and rebuilds the functional layer each cycle.

THE LAYERS OF THE UTERINE WALL, FROM THE OUTSIDE IN

Layer	What it is
Perimetrium	The outer serous layer; part of the visceral peritoneum
Myometrium	The thick middle layer of smooth muscle that contracts during childbirth
Endometrium	The inner lining; its stratum functionalis sheds in menstruation and its stratum basalis rebuilds it

Three common disorders each belong to one part of the uterus. **Uterine fibroids** are non-cancerous tumors of the smooth muscle of the myometrium. In **endometriosis**, endometrial tissue grows outside the uterus, for example on the ovaries or the pelvic peritoneum. **Cervical cancer** is a malignant tumor of the cervix.

COMPETENCY 149 LECTURE AND LAB

Vagina and external genitalia

You should be able to: Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands.

The vagina

The **vagina** is a thin-walled fibromuscular tube about 10 cm long. It runs from the cervix down and forward to the outside. It lies behind the bladder and urethra and in front of the rectum. It receives the penis, carries menstrual flow out, and serves as the birth canal. Its lining is stratified squamous epithelium in folds called **rugae**, with a layer of smooth muscle around it. It opens to the outside at the **vaginal orifice**.

The cervix pokes into the top of the vagina. The recess that rings the cervix is the **fornix**. Its posterior part, the **posterior fornix**, is the deepest. The **hymen** is a thin fold of mucous membrane at the vaginal orifice that partly covers it. Its shape and size vary a great deal from person to person.

The vulva

The external genitalia together are the **vulva**. The **mons pubis** is a rounded pad of fat over the pubic symphysis. The **labia majora** are two folds of skin and fat that run back from the mons pubis. They are covered with hair after puberty. The **labia minora** are two thin, hairless folds inside the labia majora. At the front, they meet over the clitoris and form its prepuce.

The **clitoris** is a small erectile organ at the front of the vulva, where the labia minora meet. Like the penis, it is built of erectile tissue, and only its tip, the glans of the clitoris, shows on the surface. The **vestibule** is the space between the labia minora. It holds the two main openings: the **external urethral orifice** toward the front and the vaginal orifice behind it.

The **greater vestibular glands** (Bartholin glands) are a pair of pea-sized glands, one on each side of the vaginal orifice, slightly toward the back. Their ducts open into the vestibule, and their mucus lubricates the vestibule. The **perineum** is the diamond-shaped region between the pubic symphysis and the coccyx. Its front half, the urogenital triangle, holds the external genitalia. Its back half, the anal triangle, holds the anus.

THE VAGINA AND THE STRUCTURES OF THE VULVA

Structure	Where it is	What it is
Vagina	From the cervix to the vestibule	Fibromuscular canal; the birth canal
Fornix	Top of the vagina, around the cervix	Recess that rings the cervix
Hymen	At the vaginal orifice	Thin fold of mucous membrane that partly covers the opening
Mons pubis	Over the pubic symphysis	Rounded pad of fat
Labia majora	Outer folds, back from the mons pubis	Two folds of skin and fat
Labia minora	Inside the labia majora	Two thin, hairless folds that enclose the vestibule
Clitoris	Front of the vulva, where the labia minora meet	Small erectile organ
Vestibule	Between the labia minora	Space holding the urethral and vaginal openings
Greater vestibular glands	On each side of the vaginal orifice	Glands whose mucus lubricates the vestibule

Structure	Where it is	What it is
Perineum	Between the pubic symphysis and the coccyx	Diamond-shaped region with the urogenital and anal triangles

FEMALE STRUCTURES AND THEIR MALE MATCHES

Female	Male
Ovaries	Testes
Labia majora	Scrotum
Clitoris	Penis
Greater vestibular glands	Bulbourethral glands

MEMORY TIP

Front to back along the midline: mons pubis, clitoris, external urethral orifice, vaginal orifice, then the anus behind the vestibule.

COMPETENCY 150 LECTURE AND LAB

Mammary gland structure

You should be able to: Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland.

The **mammary glands** are modified sweat glands in the breasts. Each breast lies in the subcutaneous tissue in front of the **pectoralis major** muscle. Much of the breast is adipose tissue, and the amount of fat mostly sets the size of the breast.

The gland is built in nested units. It has 15 to 20 **lobes** arranged in a ring around the nipple, separated by fat. Each lobe divides into smaller **lobules**. Each lobule holds clusters of **alveoli**, small sacs lined with cells that secrete milk. **Myoepithelial cells** around the alveoli contract to push the milk into the ducts.

Small ducts from the alveoli join into one **lactiferous duct** for each lobe. Just before the nipple, each lactiferous duct widens into a **lactiferous sinus**, where milk collects. The ducts open on the **nipple**, the raised projection near the center of the breast. The **areola** is the ring of pigmented skin around the nipple. Its small bumps are **areolar glands**, a type of sebaceous gland.

The **suspensory ligaments** (Cooper ligaments) are bands of connective tissue that run from the skin through the breast, between the lobes, to the fascia over the pectoralis major. They support the breast. Most **breast cancer** starts in the ducts. A tumor that pulls on a suspensory ligament can dimple the skin over it.

THE STRUCTURES OF THE MAMMARY GLAND

Structure	What it is
Lobes	The 15 to 20 compartments of the gland, each divided into lobules
Lobules	The subdivisions of a lobe, each holding clusters of alveoli
Alveoli	The small sacs of gland cells that secrete milk

Structure	What it is
Lactiferous ducts	One duct per lobe; carries milk toward the nipple
Lactiferous sinus	The widened end of a lactiferous duct where milk collects
Nipple	The projection where the lactiferous ducts open
Areola	The ring of pigmented skin around the nipple
Suspensory ligaments	Bands of connective tissue from the skin to the fascia over the pectoralis major that support the breast

Milk follows this route from where it is made to the outside.

1. **Alveoli** gland cells inside each lobule secrete milk
2. **Lobule ducts** small ducts carry milk out of each lobule
3. **Lactiferous duct** one duct drains each lobe toward the nipple
4. **Lactiferous sinus** the duct widens and milk collects
5. **Nipple** milk leaves through openings on the nipple

MEMORY TIP

The gland nests one way: alveoli inside lobules, lobules inside lobes. Each lobe drains by its own lactiferous duct to the nipple.

CHAPTER 4

The Brain

CENTRAL NERVOUS SYSTEM

COMPETENCIES 151 TO 160

Covers the development of the brain, the cerebrum with its lobes, white matter tracts, and basal ganglia, the functional areas of the cortex, the diencephalon, the cerebellum, the limbic system, and the arterial supply of the brain. The focus is on the structures and the job each one does.

COMPETENCIES

1. **151** Brain gross organization
2. **152** Neural tube brain development
3. **153** Cerebral surface and lobes
4. **154** Functional cortical areas
5. **155** Cerebral white matter tracts
6. **156** Basal ganglia
7. **157** Diencephalon
8. **158** Cerebellum
9. **159** Limbic system
10. **160** Cerebral arterial supply

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The brain

drsrennie-stack.github.io/new-build-bio4-solano/brain-concept-videos.html

COMPETENCY 151 LECTURE AND LAB

Brain gross organization

You should be able to: Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen.

The brain sits in the cranial cavity. At the foramen magnum it continues downward as the spinal cord. Anatomists divide the brain into four major divisions, and you can find all four on a midsagittal section of the brain model.

The **cerebrum** is the largest division. It forms the two cerebral hemispheres that fill the upper cranial cavity. The **diencephalon** sits at the core of the brain, between the hemispheres and above the brainstem. The hemispheres cover it, so you see it best on a midsagittal cut. The **brainstem** is the stalk below the diencephalon. It is made of the midbrain, pons, and medulla oblongata, and it continues as the spinal cord. The **cerebellum** sits behind the brainstem and below the occipital lobes, in the posterior cranial fossa.

THE FOUR MAJOR DIVISIONS OF THE BRAIN

Division	Where it is	Main parts
Cerebrum	Upper cranial cavity; the largest division	Two cerebral hemispheres, each with cortex, white matter, and basal nuclei
Diencephalon	Deep at the core, between the hemispheres and the brainstem	Thalamus, hypothalamus, epithalamus
Brainstem	Below the diencephalon, in front of the cerebellum; continuous with the spinal cord	Midbrain, pons, medulla oblongata

Division	Where it is	Main parts
Cerebellum	Behind the brainstem, below the occipital lobes	Two cerebellar hemispheres joined by the vermis

The brain is built from two kinds of nervous tissue. **Gray matter** holds neuron cell bodies, dendrites, unmyelinated axons, and synapses. It looks gray to tan. **White matter** is made of myelinated axons running in bundles called tracts. Myelin is fatty, so white matter looks pale.

In the cerebrum, gray matter forms a thin outer layer called the **cerebral cortex**. It is only about 2 to 4 mm thick, and it follows every gyrus and sulcus. Beneath it lies the **cerebral white matter**, the deep core of tracts. Islands of gray matter also sit deep inside the white matter. These are the **basal nuclei** and the thalamus. The cerebellum uses the same plan: gray cortex on the outside and white matter inside. The brainstem and spinal cord are arranged the other way around, with most of their white matter on the outside.

On a specimen, look at a coronal or horizontal slice. The cortex is the thin, darker rim that traces the outline of each fold. The pale tissue deep to it is white matter. Darker patches buried inside the white matter are the deep nuclei.

MEMORY TIP

Cortex is Latin for bark. Like bark on a tree, the cortex is a thin outer layer, and the white matter is the wood inside.

COMPETENCY 152 LECTURE

Neural tube brain development

You should be able to: Trace brain development from the neural tube through the primary and secondary vesicles and match each secondary vesicle to its adult structures and ventricle.

The whole central nervous system develops from the **neural tube**, which forms from ectoderm. In the early embryo a strip of ectoderm called the neural plate folds into a groove, and the edges of the groove fuse into a tube. The front (cranial) end of the tube becomes the brain. The rest of the tube becomes the spinal cord.

The tube stays hollow. Its central cavity becomes the ventricles of the brain and the central canal of the spinal cord. So as you follow each part of the tube into the adult brain, you also follow its cavity into a ventricle.

By about the fourth week, the cranial end of the tube swells into three **primary brain vesicles**: the **prosencephalon** (forebrain), the **mesencephalon** (midbrain), and the **rhombencephalon** (hindbrain).

By about the fifth week, two of these split, giving five **secondary brain vesicles**. The prosencephalon divides into the **telencephalon** and the **diencephalon**. The mesencephalon does not divide. The rhombencephalon divides into the **metencephalon** and the **myelencephalon**.

1. **Neural tube** forms from ectoderm; the cranial end becomes the brain and the rest becomes the spinal cord.
2. **Three primary vesicles** prosencephalon, mesencephalon, rhombencephalon.
3. **Five secondary vesicles** telencephalon and diencephalon (from the prosencephalon), mesencephalon (unchanged), metencephalon and myelencephalon (from the rhombencephalon).
4. **Adult brain** each secondary vesicle becomes a set of adult structures and keeps part of the tube cavity as a ventricle.

SECONDARY VESICLES, THEIR ADULT STRUCTURES, AND THEIR VENTRICLES

Secondary vesicle	Primary vesicle it came from	Adult structures	Cavity in the adult

Secondary vesicle	Primary vesicle it came from	Adult structures	Cavity in the adult
Telencephalon	Prosencephalon	Cerebrum (cerebral hemispheres, with cortex, white matter, and basal nuclei)	Lateral ventricles
Diencephalon	Prosencephalon	Thalamus, hypothalamus, epithalamus	Third ventricle
Mesencephalon	Mesencephalon	Midbrain	Cerebral aqueduct
Metencephalon	Rhombencephalon	Pons and cerebellum	Upper part of the fourth ventricle
Myelencephalon	Rhombencephalon	Medulla oblongata	Lower part of the fourth ventricle, which continues into the central canal of the spinal cord

The telencephalon grows far more than the other vesicles. The cerebral hemispheres expand backward over the diencephalon and midbrain. That growth is why the diencephalon ends up hidden deep in the adult brain.

MEMORY TIP

Read the five secondary vesicles in order from the front of the tube to the back and you name the adult brain from top to bottom: cerebrum, diencephalon, midbrain, pons and cerebellum, medulla. The prefixes help too: tel means end, di means between, mes means middle, met means after, and myel means marrow, for the part next to the spinal cord.

COMPETENCY 153 LECTURE AND LAB

Cerebral surface and lobes

You should be able to: Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen.

The cerebrum is made of two **cerebral hemispheres**, right and left. A deep midline cleft called the **longitudinal fissure** separates them. At the bottom of that fissure, the two hemispheres are joined by the corpus callosum. A second deep cleft, the **transverse fissure**, separates the back of the cerebrum from the cerebellum below it.

The surface of each hemisphere is folded. The raised ridges are **gyri** (singular, gyrus). The grooves between them are **sulci** (singular, sulcus). The deepest grooves are called **fissures**. Folding packs much more cortex into the skull than a smooth surface could hold.

A few sulci are named because they mark the lobe boundaries. The **central sulcus** runs from the top of the hemisphere down the side toward the lateral sulcus. The **precentral gyrus** lies just in front of it, bounded in front by the **precentral sulcus**. The **postcentral gyrus** lies just behind it, bounded behind by the **postcentral sulcus**. The **lateral sulcus** (Sylvian fissure) is the deep groove on the side of each hemisphere. The **parieto-occipital sulcus** is easiest to see on the medial surface, near the back of the hemisphere.

THE NAMED GROOVES OF THE CEREBRUM

Groove	Where to find it	What it separates
Longitudinal fissure	Midline, between the hemispheres	The right and left cerebral hemispheres

Groove	Where to find it	What it separates
Transverse fissure	Back of the brain, between the occipital lobes and the cerebellum	The cerebrum from the cerebellum
Central sulcus	Top and side of each hemisphere, running down toward the lateral sulcus	The frontal lobe from the parietal lobe; the precentral gyrus in front from the postcentral gyrus behind
Precentral sulcus	Just in front of the precentral gyrus	The precentral gyrus from the rest of the frontal lobe
Postcentral sulcus	Just behind the postcentral gyrus	The postcentral gyrus from the rest of the parietal lobe
Lateral sulcus (Sylvian fissure)	Deep groove on the side of each hemisphere	The temporal lobe below from the frontal and parietal lobes above
Parieto-occipital sulcus	Medial surface, near the back of the hemisphere	The parietal lobe from the occipital lobe

Each hemisphere has five lobes, so every lobe is paired, with one in each hemisphere. Four lobes are named for the cranial bone that covers them. The fifth, the **insula**, is hidden deep in the lateral sulcus. To see it, gently pull the temporal lobe away from the frontal and parietal lobes and look into the lateral sulcus.

THE FIVE LOBES AND THEIR BOUNDARIES

Lobe	Location	Boundaries
Frontal lobe	Front of each hemisphere, under the frontal bone	Central sulcus behind; lateral sulcus below
Parietal lobe	Top and upper side, under the parietal bone	Central sulcus in front; parieto-occipital sulcus behind; lateral sulcus below
Temporal lobe	Lower side, under the temporal bone	Lateral sulcus above
Occipital lobe	Back of each hemisphere, under the occipital bone	Parieto-occipital sulcus in front; transverse fissure below, separating it from the cerebellum
Insula	Deep inside the lateral sulcus	Covered by the frontal, parietal, and temporal lobes

On a specimen, find the lateral sulcus first, because it is the deepest groove on the side. Then look for the central sulcus. It usually runs without a break from the top of the hemisphere almost down to the lateral sulcus, with a long, unbroken gyrus on each side of it. Those two gyri are the precentral and postcentral gyri.

MEMORY TIP

Pre means before and post means after. The precentral gyrus comes before the central sulcus (in front of it) and the postcentral gyrus comes after it (behind it).

COMPETENCY 154 LECTURE AND LAB

Functional cortical areas

You should be able to: Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex.

The cortex is mapped into areas that each have a fixed location. There are three kinds. **Motor areas** control voluntary movement and sit in the frontal lobe. **Sensory areas** are the first cortex to receive a given kind of sensation. A primary sensory area receives that input, and an association area beside it interprets it. **Association areas** make up the rest of the cortex. They interpret input, connect it with memory, and plan responses.

Motor areas, frontal lobe

The **primary motor cortex** is the precentral gyrus. Its neurons send commands for voluntary movement to skeletal muscles on the opposite side of the body. The body is mapped along the gyrus upside down. The leg and foot are on the medial surface, inside the longitudinal fissure. The hand is higher on the lateral surface, and the face is lowest, near the lateral sulcus. Body parts with fine control, such as the hand, lips, and tongue, get the largest share of cortex. The **premotor cortex** lies just in front of the precentral gyrus and plans learned, patterned movements.

Broca's area sits in the inferior frontal gyrus, usually in the left hemisphere. It plans the movements of speech.

Sensory areas

The **primary somatosensory cortex** (primary sensory cortex) is the postcentral gyrus of the parietal lobe. It receives touch, pressure, pain, temperature, and body position from the opposite side of the body. Its body map is laid out in the same order as the motor map next to it. The somatosensory association area lies just behind it.

The special senses each have their own primary area. The **primary visual cortex** is in the occipital lobe, along the calcarine sulcus on the medial surface. The **primary auditory cortex** is on the superior temporal gyrus, along the lower edge of the lateral sulcus. The **primary gustatory cortex** (taste) is in the insula and the lower end of the postcentral gyrus that covers it. The **primary olfactory cortex** (smell) is on the medial surface of the temporal lobe, near the **uncus**. Smell reaches it from the **olfactory bulb** and **olfactory tract**, which lie on the inferior surface of the frontal lobe.

Association areas

Wernicke's area sits in the posterior part of the superior temporal gyrus, where it meets the parietal lobe, usually in the left hemisphere. It is needed to understand spoken and written language. A tract called the arcuate fasciculus connects it to Broca's area. The **prefrontal cortex** covers the front of the frontal lobe. It is a large association area that handles planning, judgment, and personality.

THE FUNCTIONAL AREAS OF THE CORTEX

Area	Kind	Lobe	Gyrus or landmark	Main job
Primary motor cortex	Motor	Frontal	Precentral gyrus	Sends commands for voluntary movement
Premotor cortex	Motor	Frontal	Just in front of the precentral gyrus	Plans learned, patterned movements
Frontal eye field	Motor	Frontal	Middle frontal gyrus	Controls voluntary eye movement
Broca's area	Motor	Frontal, usually left	Inferior frontal gyrus	Plans the movements of speech

Area	Kind	Lobe	Gyrus or landmark	Main job
Primary somatosensory cortex	Primary sensory	Parietal	Postcentral gyrus	Receives touch, pressure, pain, temperature, and body position
Somatosensory association area	Association	Parietal	Just behind the postcentral gyrus	Interprets texture, shape, and position
Primary visual cortex	Primary sensory	Occipital	Calcarine sulcus, medial surface	Receives visual input
Visual association area	Association	Occipital	Surrounding the primary visual cortex	Recognizes objects and faces
Primary auditory cortex	Primary sensory	Temporal	Superior temporal gyrus, at the lateral sulcus	Receives sound input
Auditory association area	Association	Temporal	Next to the primary auditory cortex	Interprets speech and music
Primary gustatory cortex	Primary sensory	Insula and parietal	Insula and the lower end of the postcentral gyrus	Receives taste input
Primary olfactory cortex	Primary sensory	Temporal	Medial temporal lobe, near the uncus	Receives smell input
Wernicke's area	Association	Temporal, usually left	Posterior superior temporal gyrus, next to the parietal lobe	Understanding spoken and written language
Prefrontal cortex	Association	Frontal	Front of the frontal lobe	Planning, judgment, and personality

The thalamus relays most sensory input up to these primary areas. It is a relay station deep in the diencephalon, below the cortex, and Competency 157 covers it.

MEMORY TIP

Front moves, back feels. The precentral gyrus, in front of the central sulcus, moves the body. The postcentral gyrus, behind it, feels the body.

COMPETENCY 155 LECTURE AND LAB

Cerebral white matter tracts

You should be able to: Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen.

The cerebral white matter is made of myelinated axons grouped into tracts. The tracts fall into three classes, sorted by what they connect.

Association fibers stay within one hemisphere. Short ones arc from one gyrus to the next. Long ones connect one lobe to another. The **arcuate fasciculus**, which links Broca's area and Wernicke's area, is an example.

Commissural fibers cross the midline and connect matching areas of the right and left hemispheres. The largest is the **corpus callosum**, a thick, arched band of white matter at the bottom of the longitudinal fissure. On a midsagittal section it is the C-shaped white band above the diencephalon, and it forms the roof of the lateral ventricles. The

anterior commissure and posterior commissure are smaller commissural bundles. Hanging below the corpus callosum is the **septum pellucidum**, a thin sheet that separates the two lateral ventricles. The fornix runs along its lower edge.

Projection fibers connect the cortex with the lower brain and spinal cord. Some carry signals down from the cortex, and some carry signals up to it. Just below the cortex they spread out in a fan called the **corona radiata**. Deeper down they gather into one compact band, the **internal capsule**. The internal capsule passes between the thalamus and caudate nucleus on its medial side and the lentiform nucleus (putamen and globus pallidus) on its lateral side. Below that, the descending fibers continue into the midbrain as the **crus cerebri**.

THE THREE CLASSES OF CEREBRAL WHITE MATTER FIBERS

Fiber class	What it connects	Examples	Where to see it
Association fibers	Gyri and lobes within the same hemisphere	Arcuate fasciculus (Broca area to Wernicke area)	Inside one hemisphere; they never cross the midline
Commissural fibers	Matching areas of the right and left hemispheres	Corpus callosum (largest); anterior and posterior commissures	Corpus callosum on a midsagittal section, the arched band above the diencephalon
Projection fibers	The cortex with the diencephalon, brainstem, and spinal cord, in both directions	Corona radiata; internal capsule; crus cerebri below it	Internal capsule on a coronal or horizontal section, between the thalamus and caudate nucleus and the lentiform nucleus

On a specimen, find the corpus callosum on the midsagittal cut first. To find the internal capsule, use a horizontal or coronal section. On a horizontal section it is a white band bent into a V that points toward the midline, with the caudate nucleus and thalamus on its inner side and the lentiform nucleus on its outer side.

MEMORY TIP

Commissure means a joining, so commissural fibers join the two hemispheres. Projection fibers project down to the brainstem and spinal cord or up from them. Association fibers stay in one hemisphere.

COMPETENCY 156 LECTURE AND LAB

Basal ganglia

You should be able to: Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus.

The **basal nuclei**, also called the **basal ganglia**, are paired masses of gray matter buried deep in the cerebral white matter of each hemisphere. They lie beside the lateral ventricle and the thalamus. Clusters of cell bodies in the central nervous system are usually called nuclei, so many texts use basal nuclei. Basal ganglia is the older name and is still widely used. The basal nuclei work with the motor cortex. They help start wanted movements, hold back unwanted ones, and set muscle tone.

The **caudate nucleus** is long and curved, with a head, a body, and a tail. Its large head bulges into the lateral wall of the front part (anterior horn) of the lateral ventricle. The body and tail follow the curve of the ventricle backward and then down into the temporal lobe.

The **putamen** and **globus pallidus** together form a lens-shaped mass called the **lentiform nucleus**. It sits lateral to the internal capsule. The putamen is the outer, lateral part. The globus pallidus is the inner, medial part, and it looks paler. The caudate nucleus and putamen together are called the **striatum**, because bands of gray matter cross the internal capsule between them and give the area a striped look.

Two more nuclei sit outside the cerebrum but work in the same movement circuits, so they are grouped with the basal nuclei by function. The **substantia nigra** is a band of dark, pigmented neurons in the midbrain, just behind the crus cerebri. Loss of these neurons is the change seen in Parkinson disease. The **subthalamic nucleus** is a small nucleus in the diencephalon, below the thalamus and just above the substantia nigra.

THE BASAL NUCLEI AND THE NUCLEI GROUPED WITH THEM

Structure	Brain region	Where it is	How to find it on a section
Caudate nucleus	Cerebrum	Curves along the lateral ventricle; head in the wall of the anterior horn	Gray mass bulging into the lateral ventricle, medial to the internal capsule
Putamen	Cerebrum	Lateral part of the lentiform nucleus	The outer, darker part of the lens-shaped mass lateral to the internal capsule
Globus pallidus	Cerebrum	Medial part of the lentiform nucleus	The inner, paler part of that mass, between the putamen and the internal capsule
Substantia nigra	Midbrain	Just behind the crus cerebri	Dark band across the midbrain on a cross section
Subthalamic nucleus	Diencephalon	Below the thalamus, above the substantia nigra	Small nucleus under the thalamus on a coronal section

MEMORY TIP

Pallidus means pale. The globus pallidus is the paler, inner part of the lentiform nucleus, and the putamen wraps around it on the outside.

COMPETENCY 157 LECTURE AND LAB

Diencephalon

You should be able to: Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose.

The diencephalon lies between the cerebrum and the midbrain. The cerebral hemispheres cover it, so you see it best on a midsagittal section. It has three parts, the thalamus, the hypothalamus, and the epithalamus, and together they surround the third ventricle.

The **thalamus** is a pair of egg-shaped masses of gray matter, one on each side of the third ventricle. It makes up about 80 percent of the diencephalon. The thalamus is made of many nuclei, and it relays most sensory input on its way to the cerebral cortex. The **massa intermedia** (interthalamic adhesion) is a small bridge of gray matter that crosses the third ventricle and joins the two thalami. On a midsagittal section it shows up as a small cut oval in the middle of the thalamus. Some people do not have one.

The **hypothalamus** lies below the thalamus. A shallow groove on the wall of the third ventricle, the hypothalamic sulcus, marks the border between them. The hypothalamus forms the floor and the lower walls of the third ventricle. It is the main control center for homeostasis. It controls the pituitary gland and the autonomic nervous system and helps regulate body temperature, hunger, thirst, and sleep and wake cycles. On the inferior surface of the brain, three landmarks belong to this area. The **optic chiasm** sits at its front, where the optic nerves meet and some of their fibers cross before continuing as the optic tracts. The **infundibulum** is the stalk behind the chiasm that connects the hypothalamus to the pituitary gland. The **mammillary bodies** are two small round bumps behind the infundibulum.

The **epithalamus** is the small, posterior, upper part of the diencephalon. It forms part of the roof of the third ventricle. Its main structure is the **pineal gland**, a small pine-cone-shaped endocrine gland that projects backward from the roof of the third ventricle, above the midbrain. The epithalamus also holds the habenular nuclei.

The **third ventricle** is the narrow, midline cavity enclosed by these parts. The thalami form its upper side walls, the hypothalamus forms its floor and lower walls, and the epithalamus forms the back of its roof. In front, it connects to each lateral ventricle through an interventricular foramen. Behind, it drains into the cerebral aqueduct of the midbrain.

THE PARTS OF THE DIENCEPHALON

Part	Position	Features you identify	Relation to the third ventricle
Thalamus	Upper and largest part; one on each side	Paired thalami; massa intermedia joining them	Forms the upper side walls
Hypothalamus	Below the thalamus	Optic chiasm in front, infundibulum, mammillary bodies	Forms the floor and lower side walls
Epithalamus	Back and top	Pineal gland; habenular nuclei	Forms the back of the roof
Third ventricle	Midline, between the two halves of the diencephalon	Narrow slit crossed by the massa intermedia	The cavity itself; joins the lateral ventricles in front and the cerebral aqueduct behind

MEMORY TIP

Hypo means below, so the hypothalamus sits below the thalamus. Epi means upon, so the epithalamus sits on top at the back.

COMPETENCY 158 LECTURE AND LAB

Cerebellum

You should be able to: Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections.

The cerebellum is the second largest part of the brain. It sits in the posterior cranial fossa, below the occipital lobes and behind the pons and medulla. The transverse fissure separates it from the cerebrum above. The fourth ventricle lies between the cerebellum and the brainstem. The cerebellum coordinates movement, posture, and balance, adjusting the timing and accuracy of movements without our awareness.

The cerebellum has two **cerebellar hemispheres**, joined in the midline by a narrow strip called the **vermis**. Its surface is folded into **folia**, thin, parallel, leaf-like folds that are much finer than the gyri of the cerebrum. A deep groove called the primary fissure divides the cerebellum into an anterior lobe and a posterior lobe, and a small flocculonodular lobe sits on its underside.

Inside, the cerebellum follows the same plan as the cerebrum. The **cerebellar cortex** is the outer layer of gray matter. Under it, the white matter branches out into every folium. On a midsagittal cut through the vermis this white matter looks like a branching tree, so it is called the **arbor vitae**, Latin for tree of life. Buried in the white matter are four pairs of **deep cerebellar nuclei**. From lateral to medial they are the dentate nucleus (the largest), the emboliform nucleus, the globose nucleus, and the fastigial nucleus. Most signals leaving the cerebellum start in these nuclei.

Three pairs of **cerebellar peduncles**, thick stalks of white matter, attach the cerebellum to the brainstem. Every fiber entering or leaving the cerebellum passes through one of them. Each peduncle joins a different part of the brainstem.

THE THREE PAIRS OF CEREBELLAR PEDUNCLES

Peduncle	Brainstem connection	Main traffic
Superior cerebellar peduncle	Midbrain	Mostly output, from the deep cerebellar nuclei toward the midbrain and thalamus
Middle cerebellar peduncle	Pons	Input from the pontine nuclei, which relay signals from the cerebral cortex; the largest peduncle
Inferior cerebellar peduncle	Medulla oblongata	Input from the spinal cord about body position, from the vestibular nuclei about balance, and from the olive of the medulla

On a midsagittal specimen, find the vermis cut down the middle, the arbor vitae branching through it, and the folia along its edge. The deep nuclei and the peduncles are easiest to see on models or on sections that cut through the hemispheres.

MEMORY TIP

The peduncles attach in order from top to bottom: superior to the midbrain, middle to the pons, inferior to the medulla.

COMPETENCY 159 LECTURE AND LAB

Limbic system

You should be able to: Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon.

The **limbic system** is a group of structures on the medial side of each hemisphere. Limbic comes from limbus, meaning border or ring. These structures form a ring around the diencephalon and the top of the brainstem. They work together in emotion and memory.

The **cingulate gyrus** is the gyrus that arches just above the corpus callosum on the medial surface. Below it, the **sulcus of the corpus callosum** separates it from the corpus callosum. Above it, the **cingulate sulcus** separates it from the rest of the frontal and parietal lobes. It helps connect emotion with behavior. On a midsagittal section it is the first gyrus above the corpus callosum.

The cingulate gyrus curves around the back of the corpus callosum and continues down into the temporal lobe as the **parahippocampal gyrus**, on the inferior and medial surface of the temporal lobe. The front end of the parahippocampal gyrus hooks back on itself to form the uncus. Together these gyri make the C-shaped ring that gives the system its name.

The **hippocampus** is a curved ridge of cortex rolled into the floor of the inferior horn of the lateral ventricle (also called the temporal horn), deep to the parahippocampal gyrus in the medial temporal lobe. Its name means seahorse, which describes its shape. It is needed to form new long-term memories. Its output travels in the **fornix**, an arching band of white matter that runs forward under the corpus callosum, along the lower edge of the septum pellucidum, and ends at the mammillary bodies of the hypothalamus.

The **amygdala** is an almond-shaped nucleus deep in the front of the medial temporal lobe, just in front of the hippocampus and deep to the uncus. It is involved in fear and other emotional responses, and it links emotion to memory.

THE LIMBIC STRUCTURES

Structure	Where it is	Landmark to find it	Main role
Cingulate gyrus	Medial surface of the hemisphere, arching over the corpus callosum	First gyrus above the corpus callosum on a midsagittal section, between the sulcus of the corpus callosum and the cingulate sulcus	Connects emotion with behavior
Hippocampus	Medial temporal lobe, in the floor of the inferior horn of the lateral ventricle	Deep to the parahippocampal gyrus; its output leaves in the fornix	Forms new long-term memories
Amygdala	Front of the medial temporal lobe	Deep to the uncus, just in front of the hippocampus	Fear and emotional responses; links emotion to memory
Parahippocampal gyrus	Inferior and medial surface of the temporal lobe	Continues from the cingulate gyrus; its front end forms the uncus	Completes the limbic ring
Fornix	Arch of white matter under the corpus callosum	Lower edge of the septum pellucidum on a midsagittal section	Carries hippocampal output to the mammillary bodies

MEMORY TIP

Amygdala means almond and hippocampus means seahorse. The names describe their shapes, and both sit in the medial temporal lobe, with the amygdala in front.

COMPETENCY 160 LECTURE AND LAB

Cerebral arterial supply

You should be able to: Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches.

The brain is supplied by two pairs of arteries. The two internal carotid arteries form the **anterior circulation**. The two vertebral arteries form the **posterior circulation**. The two systems join in a ring of arteries on the inferior surface of the brain, the **cerebral arterial circle**, also called the **Circle of Willis**. The ring surrounds the optic chiasm, the infundibulum, and the mammillary bodies.

Anterior circulation

Each **internal carotid artery** rises through the neck, enters the skull through the carotid canal of the temporal bone, and reaches the base of the brain beside the optic chiasm. There it divides into two branches. The **anterior cerebral artery** runs forward and medially into the longitudinal fissure and arches over the corpus callosum. It supplies the medial surface of the frontal and parietal lobes, including the leg and foot areas of the motor and somatosensory cortex. The **anterior communicating artery** is a short bridge between the right and left anterior cerebral arteries, in front of the optic chiasm. The **middle cerebral artery** is the largest branch of the internal carotid. It runs laterally into the lateral sulcus and supplies most of the lateral surface of the hemisphere. That territory includes the face and upper limb areas of the motor and somatosensory cortex, the auditory cortex, the insula, and Broca's and Wernicke's areas.

Posterior circulation

The **vertebral arteries** branch from the subclavian arteries and rise through the transverse foramina of the cervical vertebrae. They enter the skull through the foramen magnum and run up the front of the medulla. At the junction of the medulla and pons, the two vertebral arteries unite to form the **basilar artery**. The basilar artery runs up the front of the

pons in the midline, giving off small **pontine arteries** to the pons. At the top of the pons it divides into the two **posterior cerebral arteries**. Each posterior cerebral artery curves around the midbrain to supply the occipital lobe, including the visual cortex, and the inferior surface of the temporal lobe. A **posterior communicating artery** on each side joins the internal carotid artery to the posterior cerebral artery. These two vessels link the anterior and posterior circulations.

Cerebellar and spinal branches

Three pairs of cerebellar arteries come from the posterior circulation. The **posterior inferior cerebellar artery** (PICA) is the largest branch of the vertebral artery and supplies the underside of the cerebellum and the lateral medulla. The **anterior inferior cerebellar artery** (AICA) branches from the lower basilar artery and supplies the front part of the underside of the cerebellum. The **superior cerebellar artery** branches from the top of the basilar artery, just below the posterior cerebral artery, and supplies the upper surface of the cerebellum. The vertebral arteries also give rise to the spinal arteries. A small branch from each vertebral artery joins its partner in the midline to form the single **anterior spinal artery**, which runs down the front of the spinal cord. The paired **posterior spinal arteries** arise from the vertebral arteries or from the posterior inferior cerebellar arteries and run down the back of the cord.

The ring itself is made of the anterior communicating artery, the first part of each anterior cerebral artery, the end of each internal carotid artery, the two posterior communicating arteries, and the first part of each posterior cerebral artery. The middle cerebral arteries branch off the ring and are not part of it. Trace the route from a vertebral artery into the ring and around to the front:

1. **Vertebral artery** enters the skull through the foramen magnum and runs up the front of the medulla; gives off the posterior inferior cerebellar artery.
2. **Basilar artery** formed where the two vertebral arteries unite; runs up the front of the pons, giving off the anterior inferior cerebellar arteries, pontine arteries, and superior cerebellar arteries.
3. **Posterior cerebral artery** one of the two end branches of the basilar artery at the top of the pons.
4. **Posterior communicating artery** runs forward from the posterior cerebral artery to the internal carotid artery.
5. **Internal carotid artery** reaches the base of the brain beside the optic chiasm; gives off the middle cerebral artery laterally.
6. **Anterior cerebral artery** runs forward and medially toward the longitudinal fissure.
7. **Anterior communicating artery** crosses the midline to the anterior cerebral artery of the other side, closing the front of the ring.

THE ARTERIES OF THE BRAIN AND WHAT EACH SUPPLIES

Artery	Comes from	What it supplies
Internal carotid artery	Common carotid artery in the neck	Main source of the anterior circulation; ends by dividing into the anterior and middle cerebral arteries
Anterior cerebral artery	Internal carotid artery	Medial surface of the frontal and parietal lobes, including the leg and foot areas of the motor and somatosensory cortex
Anterior communicating artery	Joins the two anterior cerebral arteries	Closes the front of the ring
Middle cerebral artery	Internal carotid artery (largest branch)	Most of the lateral surface, including the face and upper limb areas of the motor and somatosensory cortex, the auditory cortex, and the insula
Posterior communicating artery	Joins the internal carotid and posterior cerebral arteries	Links the anterior and posterior circulations

Artery	Comes from	What it supplies
Posterior cerebral artery	Basilar artery (end branch)	Occipital lobe, including the visual cortex, and the inferior surface of the temporal lobe
Vertebral artery	Subclavian artery	Main source of the posterior circulation; unite to form the basilar artery
Basilar artery	Union of the two vertebral arteries	Brainstem, through the pontine arteries; gives off the anterior inferior and superior cerebellar arteries
Pontine arteries	Basilar artery	The pons
Posterior inferior cerebellar artery (PICA)	Vertebral artery	Underside of the cerebellum and the lateral medulla
Anterior inferior cerebellar artery (AICA)	Basilar artery, lower part	Front part of the underside of the cerebellum
Superior cerebellar artery	Basilar artery, top	Upper surface of the cerebellum
Anterior spinal artery	A branch from each vertebral artery, joined in the midline	Front of the spinal cord
Posterior spinal arteries	Vertebral artery or posterior inferior cerebellar artery	Back of the spinal cord

MEMORY TIP

The cerebellar arteries are named from the bottom up: PICA from the vertebral artery, AICA from the lower basilar artery, and the superior cerebellar artery from the top of the basilar artery.

LOCATION NAMES THE DEFICIT

Because each lobe and cortical area has its own job, the location of an injury predicts the loss. Damage to the **occipital lobe** affects vision. Damage to **Broca's area** makes speech slow and effortful, while understanding of language is largely spared. The artery involved predicts the pattern too. A stroke in the **middle cerebral artery** affects the face and arm on the opposite side more than the leg. A stroke in the **anterior cerebral artery** affects the opposite leg most. A clinician can often estimate where a stroke struck from the pattern of symptoms alone.

The **projection fibers** of the **internal capsule** carry most of the traffic between the cortex and the body through one narrow band. A small stroke there can cut a great deal of wiring at once, which is why such a small lesion can weaken an entire side of the body.

The **Circle of Willis** gives the brain a backup route. If one artery feeding the ring narrows slowly, such as one internal carotid artery, blood can sometimes reach its territory through the communicating arteries. This only helps for blockages in the vessels that feed the ring. A blockage in a cerebral artery beyond the ring has no backup route through the circle, and the tissue that artery feeds is at risk. In many people one of the communicating arteries is small or missing, which also limits the backup.

CHAPTER 5

The Brainstem

CENTRAL NERVOUS SYSTEM

COMPETENCIES 161 TO 166

Covers the three regions of the brainstem, the midbrain, the pons, and the medulla oblongata, the structures within each, and the reticular formation that runs through them. The focus is on the structures and the job each one does.

COMPETENCIES

1. **161** Brainstem regions
2. **162** Midbrain structures
3. **163** Pons structures
4. **164** Medulla oblongata structures
5. **165** Cranial nerve nuclei distribution
6. **166** Reticular formation

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The brainstem

drsrennie-stack.github.io/new-build-bio4-solano/brainstem-concept-videos.html

COMPETENCY 161 LECTURE AND LAB

Brainstem regions

You should be able to: Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord.

The **brainstem** is the stalk of the brain. It sits below the diencephalon, in front of the cerebellum, and above the spinal cord. It has three regions. From top to bottom they are the **midbrain**, the **pons**, and the **medulla oblongata**.

The midbrain, also called the mesencephalon, is the short, narrow region just below the diencephalon. The pons is the rounded bulge below the midbrain. Its name means bridge. The medulla oblongata is the lowest region. It tapers down and becomes the spinal cord at the **foramen magnum**.

On the front of a brain model or sheep brain, look for the groove where the pons meets the medulla. This is the **pontomedullary junction**. It is a useful landmark, because three cranial nerves (VI, VII, and VIII) leave the brainstem along it.

The brainstem connects the higher brain to the spinal cord in three ways. First, every **ascending tract** carrying sensory signals up to the brain and every **descending tract** carrying motor commands down to the cord passes through it. Second, it holds the nuclei of cranial nerves III through X and XII. The nucleus of XI sits just below it, in the upper cervical spinal cord. Third, three pairs of **cerebellar peduncles** tie the cerebellum to it: the superior peduncles to the midbrain, the middle peduncles to the pons, and the inferior peduncles to the medulla.

THE THREE REGIONS, TOP TO BOTTOM

Region	Where it is	What it connects
Midbrain	Most superior region; just below the diencephalon	Joins the diencephalon to the pons; the superior cerebellar peduncles attach here
Pons	Middle region; the rounded bulge in front of the cerebellum	Bridges the midbrain and the medulla, and links the cerebrum to the cerebellum through the middle cerebellar peduncles

Region	Where it is	What it connects
Medulla oblongata	Most inferior region; ends at the foramen magnum	Continuous with the spinal cord below; the inferior cerebellar peduncles attach here
Pontomedullary junction	The groove between the pons and the medulla on the front surface	Marks the border of the two regions; CN VI, VII, and VIII leave here

MEMORY TIP

Read the brainstem from the top down: Midbrain, then Pons, then Medulla (M, P, M). The medulla is the one that turns into the spinal cord.

COMPETENCY 162 LECTURE AND LAB

Midbrain structures

You should be able to: Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen.

The midbrain surrounds a narrow channel, the **cerebral aqueduct**. The aqueduct carries cerebrospinal fluid from the third ventricle down to the fourth ventricle. On a cross section of the midbrain it is the small hole near the back. Use it to orient the specimen: everything behind the aqueduct is the roof, and everything in front of it is the floor and the stalks.

The roof of the midbrain is the **tectum**. It carries four rounded bumps, together called the **corpora quadrigemina**. The upper pair are the **superior colliculi**. They are reflex centers for vision, such as turning the eyes and head toward a moving object. The lower pair are the **inferior colliculi**. They are reflex centers and relay stations for hearing, such as turning the head toward a sudden sound.

The front of the midbrain is formed by the two **cerebral peduncles**. These thick stalks carry the descending motor tracts from the cerebrum down to the pons, medulla, and spinal cord. The most anterior part of each peduncle is a band of white matter called the **crus cerebri**. Between the aqueduct and the crura lies the **tegmentum**, the core of the midbrain, which holds cranial nerve nuclei and part of the reticular formation.

The **substantia nigra** is a dark band of gray matter between the crus cerebri and the tegmentum. Its neurons contain a dark pigment, which is why it looks black in a fresh cross section. It works with the basal nuclei to help control movement.

Two cranial nerves leave the midbrain. The oculomotor nerve (CN III) leaves the front, in the gap between the two cerebral peduncles. The trochlear nerve (CN IV) leaves the back, just below the inferior colliculi.

THE STRUCTURES OF THE MIDBRAIN

Structure	Where it is	What it does
Tectum	The roof of the midbrain, behind the cerebral aqueduct	Carries the four colliculi
Superior colliculi	The upper pair of bumps on the tectum	Reflex centers for vision, such as tracking a moving object
Inferior colliculi	The lower pair of bumps on the tectum	Reflex centers and relays for hearing, such as turning toward a sound
Cerebral peduncles	The paired stalks on the front of the midbrain	Carry descending motor tracts from the cerebrum

Structure	Where it is	What it does
Crus cerebri	The most anterior part of each cerebral peduncle	The white matter band holding the descending motor fibers
Substantia nigra	A dark band between the crus cerebri and the tegmentum	Works with the basal nuclei to help control movement
Cerebral aqueduct	The narrow channel through the midbrain, between the tectum and the tegmentum	Carries CSF from the third ventricle to the fourth ventricle

MEMORY TIP

Superior colliculi sit higher and serve sight; inferior colliculi sit lower and serve sound. Both are reflex centers. You become aware of what you see and hear in the cerebral cortex.

COMPETENCY 163 LECTURE AND LAB

Pons structures

You should be able to: Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it.

The **pons** is the rounded bulge between the midbrain and the medulla. Its front surface is the **basilar pons**, a thick band of fibers that runs crosswise and curves back on each side toward the cerebellum. The back part of the pons is its tegmentum, which carries cranial nerve nuclei and part of the reticular formation.

Scattered through the basilar pons are the **pontine nuclei**. These clusters of gray matter receive motor information from the cerebral cortex. Their axons cross to the opposite side and enter the cerebellum. This relay is how the cerebrum keeps the cerebellum informed about planned movements.

The axons from the pontine nuclei leave the pons as the **middle cerebellar peduncles**. These are the largest of the three pairs of cerebellar peduncles. They are the thick stalks that attach the side of the pons to each cerebellar hemisphere. On a model, follow the crosswise fibers of the basilar pons around the side and into the cerebellum.

Behind the pons lies the **fourth ventricle**. It is a diamond-shaped space filled with cerebrospinal fluid. Its floor is the back of the pons and the upper medulla, and the cerebellum lies over its roof. The cerebral aqueduct opens into it from above.

The pons also holds nuclei that help adjust the breathing rhythm set by the medulla. The trigeminal nerve (CN V) leaves the side of the pons. The abducens (CN VI), facial (CN VII), and vestibulocochlear (CN VIII) nerves leave at the pontomedullary junction.

THE STRUCTURES OF THE PONS

Structure	Where it is	What it does
Basilar pons	The front surface of the pons	Crosswise fibers heading to the cerebellum, with the pontine nuclei scattered among them
Pontine nuclei	Clusters of gray matter in the basilar pons	Relay motor information from the cerebral cortex to the cerebellum
Middle cerebellar peduncles	The thick stalks on each side of the pons	Carry the axons of the pontine nuclei into the cerebellum

Structure	Where it is	What it does
Fourth ventricle	Behind the pons and upper medulla, in front of the cerebellum	Holds CSF; receives it from the cerebral aqueduct

COMPETENCY 164 LECTURE AND LAB

Medulla oblongata structures

You should be able to: Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers.

The **medulla oblongata** is the lowest region of the brainstem. It runs from the pontomedullary junction down to the foramen magnum, where it becomes the spinal cord. Its upper half forms the lower part of the floor of the fourth ventricle. Its lower half surrounds a narrow channel that continues into the cord as the central canal.

The front of the medulla has two long ridges, one on each side of the midline. These are the **pyramids**. Each pyramid is made of the **corticospinal tract**, the descending motor pathway from the cerebral cortex to the spinal cord.

Near the bottom of the medulla, most corticospinal fibers cross the midline to the opposite side. This crossing is the **decussation of the pyramids**. On a model it shows as fibers that cross over and blur the groove between the two pyramids. Because of this crossing, the left cerebral hemisphere controls the muscles on the right side of the body, and the right hemisphere controls the left side.

Just lateral to each pyramid is an oval swelling, the **olive** (the medullary olive). It is formed by the inferior olivary nucleus underneath. This nucleus relays information to the cerebellum through the inferior cerebellar peduncle. On the front of the medulla, the hypoglossal nerve (CN XII) leaves in the groove between the pyramid and the olive. The glossopharyngeal (CN IX) and vagus (CN X) nerves leave in the groove behind the olive.

The medulla also holds three **vital centers** in its core. They are part of the reticular formation and are not visible on the surface. The **cardiac center** adjusts the rate and force of the heartbeat. The **respiratory center** sets the basic rhythm of breathing. The **vasomotor center** adjusts the diameter of blood vessels and so helps set blood pressure.

THE STRUCTURES AND VITAL CENTERS OF THE MEDULLA

Structure	Where it is	What it is or does
Pyramids	Paired ridges on the front of the medulla	Formed by the descending corticospinal tracts
Decussation of the pyramids	The lower medulla, just above the spinal cord	Where most corticospinal fibers cross to the opposite side
Olive	Oval bulge lateral to each pyramid	Formed by the inferior olivary nucleus, a relay to the cerebellum
Cardiac center	Inside the medulla (not visible on the surface)	Adjusts the rate and force of the heartbeat
Respiratory center	Inside the medulla (not visible on the surface)	Sets the basic rhythm of breathing
Vasomotor center	Inside the medulla (not visible on the surface)	Adjusts blood vessel diameter, and so blood pressure

A SMALL REGION WITH VITAL JOBS

The **medulla oblongata** holds the **cardiac center**, **respiratory center**, and **vasomotor center**. Because these centers control the heartbeat and breathing, an injury to the medulla can be life-threatening in a way that an injury to a single area of cerebral cortex is not.

The **decussation of the pyramids** is the anatomical reason a stroke in the right cerebral hemisphere weakens the left side of the body. Most of the motor fibers have already crossed by the time they reach the spinal cord.

The **reticular activating system** links the brainstem to consciousness. Damage to this ascending network in the upper brainstem can cause a coma, even when the cerebral cortex itself is not injured. The **substantia nigra** links the brainstem to movement: loss of its pigmented neurons is the basis of Parkinson disease.

COMPETENCY 165 LECTURE AND LAB

Cranial nerve nuclei distribution

You should be able to: Match cranial nerve nuclei III through XII to the brainstem region that houses them.

A **cranial nerve nucleus** is a cluster of neuron cell bodies inside the brain that gives rise to the motor fibers of a cranial nerve or receives its sensory fibers. Cranial nerves I (olfactory) and II (optic) connect to the forebrain, so they have no brainstem nucleus. The nuclei of cranial nerves III through X and XII are housed in the brainstem, and the nucleus of XI sits just below it in the upper cervical spinal cord.

The nuclei line up from top to bottom in the same order as the nerve numbers. The midbrain holds III and IV. The pons holds V, VI, VII, and VIII. The medulla holds IX, X, and XII. A nerve usually leaves the brainstem near the level of its nucleus, so the attachment point on a model tells you the region.

Three nerves need a closer look. The trigeminal nerve (CN V) is assigned to the pons, where its main motor and sensory nuclei sit, but its sensory nuclei stretch up into the midbrain and down through the medulla. The vestibulocochlear nerve (CN VIII) has its vestibular and cochlear nuclei at the junction of the pons and medulla, so it is listed with both. The accessory nerve (CN XI) is grouped with the medulla in the 2–2–4–4 rule below, but its motor nucleus sits in the upper cervical spinal cord. Its rootlets rise through the foramen magnum and then leave the skull with the vagus nerve.

CRANIAL NERVE NUCLEI III TO XII BY REGION

Cranial nerve	Region that houses the nucleus	Where the nerve leaves the brainstem
III Oculomotor	Midbrain	Front of the midbrain, between the cerebral peduncles
IV Trochlear	Midbrain	Back of the midbrain, just below the inferior colliculi
V Trigeminal	Pons (sensory nuclei extend into the midbrain and medulla)	Side of the pons
VI Abducens	Pons	Pontomedullary junction, near the midline
VII Facial	Pons	Pontomedullary junction, lateral to VI
VIII Vestibulocochlear	Pons and medulla, at their junction	Pontomedullary junction, lateral to VII
IX Glossopharyngeal	Medulla	Groove behind the olive
X Vagus	Medulla	Groove behind the olive, below IX
XI Accessory	Upper cervical spinal cord (grouped with the medulla)	Side of the upper cervical cord; rootlets rise through the foramen magnum

Cranial nerve	Region that houses the nucleus	Where the nerve leaves the brainstem
XII Hypoglossal	Medulla	Groove between the pyramid and the olive

MEMORY TIP

Use the 2–2–4–4 rule. Two nerves (I and II) attach above the brainstem, two (III and IV) to the midbrain, four (V to VIII) to the pons, and four (IX to XII) to the medulla. Remember that VIII also reaches the medulla and that XI starts in the cervical cord.

COMPETENCY 166 LECTURE

Reticular formation

You should be able to: Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component.

The **reticular formation** is a loose net of gray matter that runs through the core of the whole brainstem, from the medulla up through the pons and the midbrain. Its name comes from the Latin for net. It is made of small clusters of neurons scattered among bundles of white matter, so it has no sharp borders and no surface landmark.

To locate it, look at any brainstem cross section. The reticular formation fills the central core (the tegmentum), behind the large descending motor tracts and in front of the ventricle or aqueduct. Its neurons connect with the cranial nerve nuclei, the cerebellum, the thalamus, and the spinal cord.

The reticular formation has two components that run in opposite directions. The **ascending reticular activating system**, or **RAS**, sends fibers up through the thalamus to the cerebral cortex. It keeps the cortex awake and alert, holds attention, and filters out unimportant sensory input so the cortex is not flooded. RAS activity falls during sleep. Damage to it can cause a coma.

The **descending component** sends fibers down to the spinal cord. These fibers help set the resting tone of skeletal muscles and help keep posture. The cardiac, respiratory, and vasomotor centers of the medulla are also part of the reticular formation, so this descending output links the brainstem to heart rate, breathing, and blood vessel diameter.

THE TWO COMPONENTS OF THE RETICULAR FORMATION

Component	Direction and target	What it does
Ascending reticular activating system (RAS)	Up through the thalamus to the cerebral cortex	Keeps the cortex awake and alert, holds attention, filters out unimportant sensory input
Descending component	Down to the spinal cord	Helps set resting muscle tone and posture; includes the medullary centers for heart rate, breathing, and blood vessel diameter

MEMORY TIP

Ascending means awake: the ascending system wakes up the cortex. Descending means down to the muscles: it sets their tone.

CHAPTER 6

The Meninges and Cerebrospinal Fluid

CENTRAL NERVOUS SYSTEM

COMPETENCIES 167 TO 172

Covers the cranial and spinal meninges, the dural reflections, the meningeal spaces, where the spinal cord ends and the lumbar puncture, the ventricles, cerebrospinal fluid and its circulation, and the blood-brain barrier.

COMPETENCIES

1. **167** Cranial meninges and dural reflections
2. **168** Spinal meninges and meningeal spaces
3. **169** Cord termination and lumbar puncture
4. **170** Ventricular system
5. **171** CSF production and circulation
6. **172** Blood-brain barrier structure

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Meninges and CSF

drsrennie-stack.github.io/new-build-bio4-solano/meninges-concept-videos.html

COMPETENCY 167 LECTURE AND LAB

Cranial meninges and dural reflections

You should be able to: Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein.

Three connective tissue membranes, the **meninges**, wrap the brain. From outside to inside they are the **dura mater**, the **arachnoid mater**, and the **pia mater**. Dura mater means tough mother, and pia mater means tender mother. Arachnoid means spider-like, for the web of fine strands it sends down to the pia.

Around the brain, the dura has two layers. The **periosteal layer** is attached to the inner surface of the skull. The **meningeal layer** lies inside it. The two layers are fused together except in a few places where they separate. The spaces between them form the **dural venous sinuses**, channels that collect venous blood from the brain.

The arachnoid mater is a thin, delicate membrane just inside the dura. It does not dip into the grooves of the brain. Fine strands cross from it to the pia, through the CSF-filled space between them. The pia mater is a very thin membrane rich in small blood vessels. It clings to the surface of the brain and follows it into every sulcus.

THE CRANIAL MENINGES, OUTERMOST TO INNERMOST

Layer	Position	Description
Dura mater	Outermost, against the skull	Tough; two layers (periosteal and meningeal) fused except where they part to form the dural venous sinuses
Arachnoid mater	Middle	Thin and delicate; sends web-like strands to the pia; does not follow the sulci
Pia mater	Innermost, on the brain surface	Very thin and rich in blood vessels; follows every gyrus and sulcus

The dural reflections

In three main places the meningeal layer of the dura folds inward as a sheet that divides the cranial cavity and steadies the brain. These folds are the **dural reflections**, also called the dural extensions. The **falx cerebri** drops into the longitudinal fissure between the right and left cerebral hemispheres. The **tentorium cerebelli** lies in the transverse fissure, like a tent over the cerebellum, separating it from the occipital lobes of the cerebrum above. The **falx cerebelli** is a small fold below the tentorium that separates the right and left cerebellar hemispheres.

THE DURAL REFLECTIONS

Fold	Where it is	What it separates
Falx cerebri	In the longitudinal fissure, along the midline	Right and left cerebral hemispheres
Tentorium cerebelli	In the transverse fissure, roofing the cerebellum	Cerebrum above from cerebellum below
Falx cerebelli	Midline, below the tentorium	Right and left cerebellar hemispheres

Dural venous sinus drainage

The main sinuses run in the edges of these folds. The **superior sagittal sinus** runs along the top edge of the falx cerebri, where it attaches to the skull. The **inferior sagittal sinus** runs in the lower free edge of the falx cerebri. It drains into the **straight sinus**, which runs where the falx cerebri meets the tentorium cerebelli. The superior sagittal and straight sinuses meet at the back of the skull, at the **confluence of sinuses**. Follow the blood out of the skull from the superior sagittal sinus.

1. **Superior sagittal sinus** runs from front to back along the top edge of the falx cerebri; the arachnoid granulations return CSF into it
2. **Confluence of sinuses** the meeting point at the back of the skull, at the internal occipital protuberance; the straight sinus also drains here
3. **Transverse sinuses** paired sinuses that run laterally from the confluence along the back edge of the tentorium cerebelli
4. **Sigmoid sinuses** S-shaped sinuses that curve down from each transverse sinus to the jugular foramen
5. **Internal jugular veins** the sigmoid sinus passes through the jugular foramen and becomes the internal jugular vein, which carries the blood down the neck toward the heart

COMPETENCY 168 LECTURE AND LAB

Spinal meninges and meningeal spaces

You should be able to: Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents.

The same three meninges continue down the vertebral canal around the spinal cord. The **spinal dura mater** is a single layer. It matches the meningeal layer of the cranial dura and is not attached to the vertebrae. It forms a tube, the dural sac, that ends at about the level of S2. The **spinal arachnoid mater** lines the dural sac. The **spinal pia mater** clings to the surface of the cord.

Two extensions of the pia mater hold the cord in place. The **denticulate ligaments** are tooth-shaped flaps of pia along each side of the cord, between the dorsal and ventral roots. They reach out through the arachnoid and attach to the dura, which keeps the cord centered in the dural sac. The **filum terminale** is a thin strand of pia that continues down from the bottom tip of the cord. It passes through the lumbar cistern, picks up a covering of dura at the end of the dural sac, and anchors to the coccyx.

The meningeal spaces

The **epidural space** lies between the dura and the bone. In the vertebral canal it is a real space, filled with fat and veins, and it is where an epidural injection goes. In the skull the periosteal dura is attached to the bone, so the cranial epidural space is only a potential space. It opens only if something, such as blood, separates the dura from the skull.

The **subdural space** lies between the dura and the arachnoid. Normally the two membranes touch, so it is a potential space, holding only a thin film of fluid. The **subarachnoid space** lies between the arachnoid and the pia. It is a real space. It holds cerebrospinal fluid, the web-like strands of the arachnoid, and the larger blood vessels that run over the surface of the brain and cord.

THE THREE MENINGEAL SPACES

Space	Location	Contents
Epidural space	Between the dura and the bone	Spinal: fat and veins (a real space). Cranial: a potential space only
Subdural space	Between the dura and the arachnoid	A potential space with a thin film of fluid
Subarachnoid space	Between the arachnoid and the pia	Cerebrospinal fluid, arachnoid strands, and the larger surface blood vessels

Bleeding and infection are named by the layer or space they involve.

DISORDERS NAMED BY MENINGEAL LAYER OR SPACE

Disorder	Structure affected	What changes there
Epidural hematoma	Between the skull and the dura	Blood, often from an artery torn by a skull fracture, separates the dura from the bone
Subdural hematoma	Subdural space	Blood, often from torn veins crossing to the dural sinuses, collects between the dura and the arachnoid
Subarachnoid hemorrhage	Subarachnoid space	Blood from a ruptured surface artery mixes into the cerebrospinal fluid
Meningitis	The meninges, mainly the arachnoid and pia	The membranes become inflamed, usually from an infection; the CSF in the subarachnoid space shows the change

COMPETENCY 169 LECTURE AND LAB

Cord termination and lumbar puncture

You should be able to: Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space.

The spinal cord is shorter than the vertebral column. In an adult it ends at about the level of the disc between L1 and L2. The tapered, cone-shaped end of the cord is the **conus medullaris**.

The nerve roots for the lumbar, sacral, and coccygeal spinal nerves leave the cord above that level. They have to travel down the vertebral canal before they reach their exit openings. This bundle of long roots below the conus medullaris is the **cauda equina**, which means horse's tail.

The dural sac and the subarachnoid space continue down to about S2. Below the conus medullaris the subarachnoid space is wide and holds no spinal cord, only CSF, the roots of the cauda equina, and the filum terminale. This enlarged space is the **lumbar cistern**.

THE END OF THE CORD AND WHAT LIES BELOW IT

Structure	Level	What it is
Conus medullaris	L1 to L2 disc (adult)	The tapered lower end of the spinal cord

Structure	Level	What it is
Cauda equina	Below L1 to L2, down the vertebral canal	The bundle of lumbar, sacral, and coccygeal nerve roots
Filum terminale	From the conus medullaris to the coccyx	A strand of pia mater that anchors the lower end of the cord
Lumbar cistern	From the conus medullaris to about S2	The enlarged subarachnoid space, filled with CSF and the cauda equina

The lumbar puncture

A **lumbar puncture**, or spinal tap, draws CSF from the lumbar cistern. The needle goes in at the midline between the spinous processes of L3 and L4 or L4 and L5, well below the end of the cord. A line drawn across the tops of the iliac crests crosses the spine at about L4, which helps find the level. The roots of the cauda equina float in the CSF and slide aside from the needle. List the layers the needle crosses, superficial to deep.

1. **Skin** the epidermis and dermis of the lower back
2. **Subcutaneous tissue** the fatty layer under the skin
3. **Supraspinous ligament** runs along the tips of the spinous processes
4. **Interspinous ligament** fills the gap between neighboring spinous processes
5. **Ligamentum flavum** the tough, elastic ligament joining the laminae of neighboring vertebrae; the needle meets firm resistance here and then a give
6. **Epidural space** the fat-filled and vein-filled space between the bone and the dura
7. **Dura mater** the tough outer meningeal membrane
8. **Arachnoid mater** the thin middle membrane, pressed against the dura, so the needle passes the two together
9. **Subarachnoid space** the needle tip is now in the lumbar cistern, and CSF can be drawn

THE CORD ENDS, BUT THE FLUID DOES NOT

The spinal cord stops at the **conus medullaris** near the L1 to L2 disc, yet the subarachnoid space runs lower, as the **lumbar cistern**. This difference in length is the basis of the lumbar puncture. A needle placed between the L3 and L4 or L4 and L5 spines reaches a pool of CSF where only the floating **cauda equina** lies in its path.

Knowing the layers in order is what makes the tap safe. The needle crosses skin, subcutaneous tissue, three ligaments, the **epidural space**, and then the dura and arachnoid before CSF appears. The firm **ligamentum flavum** giving way is a landmark the clinician feels.

The CSF drawn this way is tested for signs of meningitis or bleeding. The named spaces also tell bleeds apart: an **epidural hematoma** lies between the skull and the dura, a **subdural hematoma** lies just beneath the dura, and a **subarachnoid hemorrhage** mixes into the CSF itself.

COMPETENCY 170 LECTURE AND LAB

Ventricular system

You should be able to: Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them.

The **ventricles** are connected, fluid-filled spaces inside the brain. There are four: two lateral ventricles, the third ventricle, and the fourth ventricle. They are lined by **ependymal cells**, a type of glial cell, and filled with cerebrospinal fluid. Below the fourth ventricle the same system continues as the central canal of the spinal cord.

The two **lateral ventricles** are the largest. There is one in each cerebral hemisphere. Each is C-shaped, curving from the frontal lobe back and then down into the temporal lobe. In the midline the two lateral ventricles are separated by a thin sheet called the **septum pellucidum**.

The **third ventricle** is a narrow slit in the midline of the diencephalon, between the right and left halves of the thalamus. Each lateral ventricle opens into it through a small hole, the **interventricular foramen** (foramen of Monro). There is one foramen on each side.

The third ventricle drains down through the **cerebral aqueduct** (aqueduct of Sylvius), the narrow channel through the midbrain. The aqueduct opens into the **fourth ventricle**, which lies behind the pons and upper medulla and in front of the cerebellum.

The fourth ventricle has two kinds of exits. Three apertures in its roof and sides open into the subarachnoid space around the brain: one **median aperture** (foramen of Magendie) in the midline and two **lateral apertures** (foramina of Luschka), one on each side. Separately, the bottom of the fourth ventricle narrows directly into the **central canal**, which runs down the center of the spinal cord. The apertures lead only to the subarachnoid space, and the central canal opens only from the fourth ventricle itself.

THE VENTRICLES AND HOW THEY CONNECT

Space	Location	Connects to	Through
Lateral ventricles	One in each cerebral hemisphere; separated by the septum pellucidum	Third ventricle	Interventricular foramina (of Monro)
Third ventricle	Midline of the diencephalon, between the halves of the thalamus	Fourth ventricle	Cerebral aqueduct (of Sylvius)
Fourth ventricle	Behind the pons and upper medulla, in front of the cerebellum	Subarachnoid space	One median aperture and two lateral apertures
Fourth ventricle	Lower end, in the medulla	Central canal	Continues directly down into the cord
Central canal	Center of the spinal cord, full length	Fourth ventricle above	Direct continuation

MEMORY TIP

Count the openings out of the fourth ventricle into the subarachnoid space: one median and two lateral, three in all. The central canal is a straight continuation down into the cord.

COMPETENCY 171 LECTURE AND LAB

CSF production and circulation

You should be able to: Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations.

Cerebrospinal fluid (CSF) is a clear, watery fluid that fills the ventricles and central canal and surrounds the brain and spinal cord in the subarachnoid space. It floats the brain, which lowers the brain's effective weight. It cushions the brain and cord against jolts. It also keeps a stable chemical environment around the neurons.

CSF is made by the **choroid plexus**. A choroid plexus is a tuft of capillaries covered by ependymal cells that hangs into a ventricle. The ependymal cells make CSF from the blood plasma in the capillaries. There is choroid plexus in all four ventricles, and the largest amounts are in the lateral ventricles. About 150 mL of CSF surrounds the CNS at any one time, and about 500 mL is made each day, so the fluid is replaced several times a day.

CSF moves one way through the ventricles, out into the subarachnoid space, and back into the blood. It is reabsorbed at the **arachnoid granulations**, small knobs of arachnoid mater that push through the dura into the dural venous sinuses, mainly the superior sagittal sinus. From there the fluid leaves the skull with the venous blood. Follow the route in order.

1. **Choroid plexus** capillaries covered by ependymal cells make CSF in each ventricle, most of it in the lateral ventricles
2. **Lateral ventricles** CSF collects in the ventricle in each cerebral hemisphere
3. **Interventricular foramina** CSF passes through these paired openings (foramina of Monro) into the third ventricle
4. **Third ventricle** in the midline of the diencephalon; its choroid plexus adds more CSF
5. **Cerebral aqueduct** CSF flows down this narrow channel through the midbrain
6. **Fourth ventricle** behind the pons and upper medulla; its choroid plexus adds more CSF. A small amount continues straight down into the central canal of the spinal cord
7. **Median and lateral apertures** CSF leaves the fourth ventricle through one median aperture and two lateral apertures
8. **Subarachnoid space** CSF bathes the outside of the brain and spinal cord, including the lumbar cistern
9. **Arachnoid granulations** CSF passes through these knobs of arachnoid mater into the venous blood
10. **Dural venous sinuses** the CSF now travels with the venous blood, mainly in the superior sagittal sinus, and drains out of the skull through the internal jugular veins

WHAT CSF IS, WHERE IT IS, AND WHAT IT DOES

Feature	Description
Where CSF is found	The four ventricles, the central canal, and the subarachnoid space around the brain and cord, including the lumbar cistern
Site of production	The choroid plexus in the lateral, third, and fourth ventricles
Site of reabsorption	The arachnoid granulations, into the dural venous sinuses
Buoyancy	Floats the brain so it does not press down on its own base
Protection	Cushions the brain and spinal cord against jolts and blows
Chemical stability	Carries nutrients, removes wastes, and keeps a steady environment around the neurons

The narrow points on this route are the interventricular foramina, the cerebral aqueduct, and the apertures of the fourth ventricle. If CSF is blocked at one of them, or is not reabsorbed, it builds up. This condition is **hydrocephalus**.

A DISORDER OF CSF CIRCULATION

Disorder	Structure affected	What changes there
Hydrocephalus	The ventricles	CSF builds up behind a blockage or from poor reabsorption; the ventricles enlarge and pressure inside the skull often rises

MEMORY TIP

CSF is made inside the brain, in the ventricles, and reabsorbed outside the brain, at the arachnoid granulations in the dural sinuses. Anything that blocks a narrow point between those two places backs the fluid up.

COMPETENCY 172 LECTURE AND LAB

Blood-brain barrier structure

You should be able to: Name the three structural components of the blood-brain barrier the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet.

The **blood-brain barrier** (BBB) controls what can pass from the blood into the brain tissue. It keeps out many substances, including many drugs and germs, while letting in what neurons need, such as glucose and oxygen. It is a structure built around the brain's capillaries, with three components.

The first and most important component is the **capillary endothelium**. The endothelial cells that line brain capillaries are joined by **tight junctions**. These seal the gaps between neighboring cells. Brain capillaries also lack the pores and wide clefts found in many capillaries elsewhere in the body. As a result, most substances can only cross by passing through the endothelial cells themselves.

The second component is the **basement membrane**, a thin layer of extracellular material that surrounds the endothelium. The third is the **astrocyte end-feet**. Astrocytes are star-shaped glial cells. Their processes end in flattened feet that wrap around the outside of the capillary, covering the basement membrane. The end-feet support the endothelial cells and help keep the tight junctions in place.

The barrier is missing in a few small areas around the ventricles, such as parts of the hypothalamus, where the brain needs to sample the blood directly.

THE THREE COMPONENTS OF THE BLOOD-BRAIN BARRIER, FROM THE BLOOD OUTWARD

Component	What it is	Its part in the barrier
Capillary endothelium	The cells lining the brain capillaries	Joined by tight junctions and lacking pores, so substances cannot slip between the cells
Basement membrane	A thin layer of extracellular material around the endothelium	Surrounds and supports the capillary wall
Astrocyte end-feet	The flattened ends of astrocyte processes	Wrap the outside of the capillary and help maintain the tight junctions

MEMORY TIP

Go from the blood outward: endothelium, then basement membrane, then astrocyte feet. The tight junctions in the endothelium are the actual seal.

CHAPTER 7

Functional Organization and Nervous Tissue

NERVOUS TISSUE

COMPETENCIES 173 TO 178

Covers the organization of the nervous system and its divisions, the three functions, the structure of a neuron, the synapse, the classification of neurons, the neuroglia, myelination, and the collections of nervous tissue. The focus is on the structures and the job each one does.

COMPETENCIES

1. **173** Nervous system divisions
2. **174** Neuron structure
3. **175** Synapse structures
4. **176** Neuron classification
5. **177** Neuroglia identification
6. **178** Myelination and regeneration

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Nervous tissue

drsrennie-stack.github.io/new-build-bio4-solano/nervous-tissue-concept-videos.html

COMPETENCY 173 LECTURE

Nervous system divisions

You should be able to: Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions.

The nervous system is the body's fast control system. It senses changes, decides what they mean, and sends out commands. Anatomists divide it in two ways: by where the structures sit, and by what the motor neurons control.

The **central nervous system (CNS)** is the **brain** and the **spinal cord**. The brain sits in the cranial cavity. The spinal cord runs down the vertebral canal and links the brain to the body. The CNS is where information is processed and where commands are issued.

The **peripheral nervous system (PNS)** is every neural structure outside the brain and spinal cord. It includes the **nerves**, the **ganglia**, and the **sensory receptors**. There are 12 pairs of **cranial nerves**, which attach to the brain, and 31 pairs of **spinal nerves**, which attach to the spinal cord. The PNS carries information between the CNS and every other part of the body.

The PNS has a sensory side and a motor side. The **sensory (afferent) division** carries signals from receptors toward the CNS. The **motor (efferent) division** carries commands away from the CNS to the **effectors**, the muscles and glands that respond.

The motor side is split by the effector it controls. The **somatic nervous system** sends commands to skeletal muscle, and you control it voluntarily. The **autonomic nervous system (ANS)** sends commands to smooth muscle, cardiac muscle, and glands, and it works without conscious control. The ANS has two divisions: the **sympathetic division** for fight or flight, and the **parasympathetic division** for rest and digest. Many organs receive both.

The **enteric nervous system (ENS)** is a network of roughly 100 million neurons in the wall of the digestive tract. It controls the smooth muscle and glands of the gut and can run on its own, though the ANS connects to it and adjusts it. Some textbooks count it as part of the ANS.

The outline

- Nervous system
  - Central nervous system (CNS): brain and spinal cord
  - Peripheral nervous system (PNS): nerves, ganglia, sensory receptors
    - Sensory (afferent) division: toward the CNS
    - Motor (efferent) division: away from the CNS
      - Somatic nervous system: skeletal muscle
      - Autonomic nervous system: smooth muscle, cardiac muscle, glands (sympathetic and parasympathetic divisions)
      - Enteric nervous system: the wall of the digestive tract

THE TWO DIVISIONS AND THE STRUCTURES IN EACH

Structure	What it is
Central nervous system (CNS)	The brain and the spinal cord; it processes information and issues commands
Brain	The CNS organ in the cranial cavity
Spinal cord	The CNS cord in the vertebral canal that links the brain to the body
Peripheral nervous system (PNS)	Every neural structure outside the brain and spinal cord
Nerves	Bundles of axons in the PNS; 12 pairs of cranial nerves and 31 pairs of spinal nerves
Ganglia	Small clusters of neuron cell bodies outside the CNS
Sensory receptors	Structures that detect changes inside and outside the body

THE MOTOR DIVISIONS OF THE PNS

Division	Effector it controls	Control
Somatic nervous system	Skeletal muscle	Voluntary
Autonomic nervous system (ANS)	Smooth muscle, cardiac muscle, and glands	Involuntary
Sympathetic division (ANS)	The same ANS effectors, for exercise and emergencies (fight or flight)	Involuntary
Parasympathetic division (ANS)	The same ANS effectors, for routine activities (rest and digest)	Involuntary
Enteric nervous system (ENS)	Smooth muscle and glands in the wall of the digestive tract	Involuntary; can work on its own

The three functions

Every task the nervous system performs runs through the same three steps.

1. **Sensory function**: receptors detect a change and send the input toward the CNS.
2. **Integrative function**: the CNS analyzes the input, stores some of it, and decides on a response.
3. **Motor function**: the CNS sends commands out to the effectors, which carry out the response.

MEMORY TIP

Afferent arrives at the CNS. Efferent exits the CNS. The motor divisions are named for what they run: somatic for skeletal muscle, autonomic for the organs, enteric for the gut.

COMPETENCY 174 LECTURE AND LAB

Neuron structure

You should be able to: Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them.

A **neuron** is the nerve cell that carries signals. Mature neurons are excitable, they can live for a whole lifetime, and they generally do not divide. Every neuron has a cell body and two kinds of processes: dendrites and an axon.

The **cell body** (also called the soma or perikaryon) holds the **nucleus**, which has a large, dark **nucleolus**, and most of the cytoplasm and organelles. It is the metabolic center of the neuron and makes the proteins the whole cell needs. Scattered through its cytoplasm are **Nissl bodies**, clusters of rough endoplasmic reticulum and free ribosomes where those proteins are made. Nissl bodies stain dark on a slide and appear in the cell body and the dendrites. **Neurofibrils**, bundles of cytoskeleton filaments, run through the cell and give it its shape.

Dendrites are the short, heavily branched processes that receive incoming signals. They carry those signals toward the cell body. A multipolar neuron, the kind you see on the lab slide and the neuron model, has many dendrites.

The **axon hillock** is the cone-shaped region where the cell body narrows into the axon. It has no Nissl bodies, so it stains paler than the rest of the cell body. That pale cone is how you find the axon on a slide and tell it apart from a dendrite. Nerve impulses begin at the trigger zone, where the axon hillock meets the first part of the axon.

The **axon** is the single long process that carries the impulse away from the cell body. Its cytoplasm is the **axoplasm** and its plasma membrane is the **axolemma**. An axon can give off side branches called **axon collaterals**. At its far end the axon divides into many fine branches, the **axon terminals**, and the tip of each terminal swells into a **synaptic end bulb** that contacts the next cell.

The direction a signal travels

1. **Dendrites** receive the incoming signal and carry it toward the cell body
2. **Cell body** collects the signals arriving from the dendrites
3. **Axon hillock** the impulse starts at the trigger zone where the hillock joins the axon
4. **Axon** carries the impulse away from the cell body
5. **Axon terminals** branch at the end of the axon and end in synaptic end bulbs that pass the signal to the next cell

THE PARTS OF A NEURON

Structure	What it is
Cell body	The soma or perikaryon; it holds the nucleus and cytoplasm and is the metabolic center of the neuron
Nucleus and nucleolus	The cell's nucleus, with a large, dark-staining nucleolus, inside the cell body
Nissl bodies	Clusters of rough ER and ribosomes in the cell body and dendrites; the site of protein synthesis
Neurofibrils	Bundles of cytoskeleton filaments that give the neuron its shape and support
Microtubules	Cytoskeletal tubes that transport materials along the neuron

Structure	What it is
Dendrites	The branched processes that receive incoming signals and carry them toward the cell body
Axon hillock	The cone-shaped region where the cell body joins the axon; it has no Nissl bodies
Axon	The single long process that carries the impulse away from the cell body
Axoplasm and axolemma	The cytoplasm of the axon and the plasma membrane of the axon
Axon terminals	The fine branches at the far end of the axon that end in synaptic end bulbs
Lipofuscin	A yellow-brown pigment that builds up in neurons with age; it does not appear to harm the cell

MEMORY TIP

Dendrites deliver to the cell body. The axon carries the signal away. On a slide, the pale cone with no Nissl bodies marks the start of the axon.

COMPETENCY 175 LECTURE AND LAB

Synapse structures

You should be able to: Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact.

A **synapse** is the junction where a neuron passes its signal to another cell. The cell that sends the signal is the **presynaptic neuron**. The cell that receives it is the **postsynaptic cell**, which can be another neuron, a muscle fiber, or a gland cell. The synapse between a motor neuron and a skeletal muscle fiber is the neuromuscular junction.

At a **chemical synapse** the two cells do not touch. A narrow gap, the **synaptic cleft**, separates them. It is only about 20 to 50 nanometers wide. The signal crosses this gap as a chemical, a **neurotransmitter**. The other kind of synapse, an electrical synapse, joins two cells directly through gap junctions and is much less common.

On the sending side, each axon terminal ends in a swelling called a **synaptic end bulb**. Your lab list calls it the synaptic bulb (terminal end). Inside the end bulb are many **synaptic vesicles**, tiny membrane sacs that store neurotransmitter. The membrane of the end bulb that faces the cleft is the presynaptic membrane. Across the cleft, the **postsynaptic membrane** holds the receptors that the neurotransmitter binds to.

Only the presynaptic side has vesicles, so a chemical synapse passes its signal in one direction only.

1. **Synaptic end bulb** the nerve impulse arrives at the end of the axon terminal
2. **Synaptic vesicles** fuse with the presynaptic membrane and release neurotransmitter
3. **Synaptic cleft** the neurotransmitter diffuses across the gap
4. **Postsynaptic membrane** the neurotransmitter binds receptors, and the signal passes to the next cell

Three synapse types, named for the site the axon contacts

Each name starts with "axo" for the presynaptic axon and ends with the part of the next neuron it touches. An **axodendritic synapse** is an axon ending on a dendrite, and it is the most common type. An **axosomatic synapse** is an axon ending on a cell body (soma). An **axoaxonic synapse** is an axon ending on another axon.

THE SYNAPSE AND ITS PARTS

Structure	What it is

Structure	What it is
Synapse	The site where a neuron communicates with another neuron, a muscle cell, or a gland
Chemical synapse	A synapse with no direct contact; the signal crosses as neurotransmitter molecules
Synaptic end bulbs	The swellings at the tips of the axon terminals that hold the synaptic vesicles
Synaptic vesicles	The tiny sacs inside the end bulbs that store neurotransmitter
Synaptic cleft	The narrow gap between the two cells that the neurotransmitter diffuses across
Postsynaptic membrane	The membrane of the receiving cell, which holds the neurotransmitter receptors
Neurotransmitters	The chemical messengers that carry the signal across the cleft to the next cell
Axodendritic synapse	An axon ending on a dendrite
Axosomatic synapse	An axon ending on a cell body
Axoaxonic synapse	An axon ending on another axon

MEMORY TIP

Read the synapse names left to right: the first half is always the axon, and the second half is where it lands. Dendrite, soma, or axon.

COMPETENCY 176 LECTURE AND LAB

Neuron classification

You should be able to: Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each.

Neurons are classified in two ways. The structural classification counts how many processes leave the cell body. The functional classification asks which direction the neuron carries its signal, toward the CNS or away from it.

Structural classification, by number of processes

A **multipolar neuron** has many dendrites and one axon. It is the most common type in the body. Motor neurons and interneurons are multipolar, and so is the neuron on your lab slide. Two multipolar neurons you will see on slides are the **pyramidal cells** of the cerebral cortex and the **Purkinje cells** of the cerebellum.

A **bipolar neuron** has one dendrite and one axon, leaving from opposite ends of the cell body. Bipolar neurons are rare. They are found in special sense organs: the retina of the eye, the olfactory epithelium of the nose, and the inner ear.

A **unipolar neuron** has one short process that leaves the cell body and then splits like the letter T into two branches. Some textbooks call it pseudounipolar. The peripheral branch starts at a receptor, and the central branch enters the CNS. Most sensory neurons are unipolar, and their cell bodies sit in the dorsal root ganglia and the cranial nerve ganglia.

NEURONS NAMED FOR THE NUMBER OF PROCESSES

Type	Processes	Example
Multipolar neuron	Many dendrites and one axon; the most common type	A motor neuron in the anterior horn of the spinal cord; pyramidal cells; Purkinje cells
Bipolar neuron	One dendrite and one axon	Neurons in the retina, the olfactory epithelium, and the inner ear

Type	Processes	Example
Unipolar neuron	One process that splits into a peripheral branch and a central branch	A touch or pain sensory neuron with its cell body in a dorsal root ganglion
Pyramidal cells	A large multipolar neuron with a pyramid-shaped cell body	The cerebral cortex
Purkinje cells	A large multipolar neuron with a fan of dendrites	The cerebellar cortex

Functional classification, by direction

Sensory (afferent) neurons carry signals from receptors toward the CNS. A neuron that carries touch from the skin of a fingertip to the spinal cord is a sensory neuron. Most are unipolar.

Motor (efferent) neurons carry commands away from the CNS to an effector. A motor neuron whose cell body sits in the anterior horn of the spinal cord and whose axon runs to the biceps brachii is an example. Motor neurons are multipolar.

Interneurons, also called association neurons, lie entirely within the CNS. They connect sensory neurons to motor neurons and to each other. The neurons in the spinal cord gray matter that link the sensory and motor neurons of a withdrawal reflex are interneurons. Most neurons in the body are interneurons, and they are multipolar.

NEURONS NAMED FOR THE DIRECTION THEY CARRY A SIGNAL

Type	Direction	Example
Sensory (afferent) neurons	Toward the CNS; most are unipolar	A touch neuron from the fingertip to the spinal cord
Motor (efferent) neurons	Away from the CNS; multipolar	An anterior horn motor neuron to the biceps brachii
Interneurons	Within the CNS, between other neurons; multipolar	Spinal cord neurons linking the sensory and motor neurons of a withdrawal reflex

MEMORY TIP

SAME: Sensory is Afferent, Motor is Efferent. For shape, count the processes leaving the cell body: many, two, or one.

COMPETENCY 177 LECTURE AND LAB

Neuroglia identification

You should be able to: Name the four CNS glia and two PNS glia and describe the support role each one performs.

Neuroglia, or glial cells, are the support cells of nervous tissue. They are smaller than neurons and very numerous, though textbooks differ on how many glia there are for each neuron. Unlike mature neurons, they can still divide. In the CNS they make up about half the volume of the tissue. On a slide, the many small dark nuclei around the large neuron cell bodies belong to glia.

There are six types. Four are in the CNS and two are in the PNS.

The four glia of the CNS

Astrocytes are star-shaped cells and the most numerous glia in the CNS. Their processes wrap around neurons and capillaries. They hold neurons in place, help form the **blood-brain barrier**, and keep the fluid around neurons at the right chemical balance. In the embryo they guide growing neurons, and after an injury they fill the damaged area with scar tissue.

Oligodendrocytes have a few long processes. Each process wraps around a segment of an axon to form myelin, so one oligodendrocyte myelinates parts of several different axons. They are the myelin-forming cells of the CNS.

Microglia are small cells that act as phagocytes. They engulf microbes and clear away debris from dead and damaged cells. They are the CNS's own immune defense.

Ependymal cells are cuboidal to columnar cells, many with cilia, that line the ventricles of the brain and the central canal of the spinal cord. In the choroid plexuses they help produce cerebrospinal fluid, and their cilia help it circulate.

The two glia of the PNS

Schwann cells wrap around PNS axons. Each Schwann cell forms the myelin around one segment of a single axon. Schwann cells also enclose unmyelinated PNS axons, and they guide a damaged axon as it regrows.

Satellite cells are flat cells that surround the cell bodies of neurons in ganglia. They support those cell bodies and regulate the exchange of materials between the cell bodies and the fluid around them.

THE FOUR GLIA OF THE CNS AND WHAT EACH ONE DOES

Glial cell	What it looks like	Support role
Astrocytes	Star-shaped; the most numerous CNS glia	Support neurons, help form the blood-brain barrier, keep the chemical environment around neurons stable, guide neuron growth, form scar tissue after injury
Oligodendrocytes	A few long processes that wrap axons	Form the myelin sheaths of the CNS, each cell on several axons
Microglia	Small cells	Phagocytes that remove microbes and debris
Ependymal cells	Cuboidal to columnar, many with cilia	Line the ventricles and central canal; help produce and circulate cerebrospinal fluid

THE TWO GLIA OF THE PNS AND WHAT EACH ONE DOES

Glial cell	What it looks like	Support role
Schwann cells	Wrapped around a segment of one axon	Form the myelin sheath around PNS axons; aid axon regeneration
Satellite cells	Flat cells around neuron cell bodies	Support the cell bodies of PNS neurons in ganglia and regulate their surroundings

MEMORY TIP

The two PNS glia both start with S: Schwann and Satellite. Every other glial cell is in the CNS.

COMPETENCY 178 LECTURE AND LAB

Myelination and regeneration

You should be able to: Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential.

The **myelin sheath** is a covering made of many layers of a glial cell's plasma membrane, wrapped tightly around an axon. Because it is mostly lipid with some protein, it looks white. It insulates the axon and makes the nerve impulse travel much faster. An axon with this covering is a myelinated axon.

Myelin is laid down in segments along the axon. The short gaps between neighboring segments are the **nodes of Ranvier**, where the axolemma is left bare. The myelin-covered stretch between two nodes is an **internode**. In a myelinated axon the impulse jumps from node to node. This fast form of conduction is called **saltatory conduction**.

The **neurolemma** is the outermost layer of a Schwann cell. It holds the Schwann cell's nucleus and most of its cytoplasm, and it sits outside the myelin layers. Only Schwann cells have one, so the neurolemma is found only around PNS axons.

PNS and CNS myelination compared

In the PNS, the myelin is made by **Schwann cells**. Each Schwann cell wraps one segment of one axon, so a long axon needs a row of Schwann cells along its length. In the CNS, the myelin is made by **oligodendrocytes**. One oligodendrocyte sends processes to several axons and myelinates a segment of each. Oligodendrocytes have no neurolemma.

That difference decides whether a cut axon can regrow. When a PNS axon is cut, the part beyond the cut breaks down. If the cell body is intact, the Schwann cells line up and their neurolemma forms a **regeneration tube** that guides the regrowing axon back toward its target. In the CNS there is no neurolemma to guide regrowth. Oligodendrocytes also release substances that block regrowth, and astrocytes fill the area with scar tissue. Regeneration in the CNS is very limited.

Myelination increases from birth to maturity, so an infant's nerve conduction is slower than an adult's.

THE MYELIN SHEATH AND ITS PARTS

Structure	What it is
Myelin sheath	A multilayered covering of lipid and protein wrapped around an axon; it insulates the axon and speeds up the impulse
Nodes of Ranvier	The gaps in the myelin sheath, spaced along the axon
Internode	The myelin-covered segment between two nodes
Saltatory conduction	The fast conduction in which the impulse jumps from node to node
Neurolemma	The outer layer of a Schwann cell, holding its nucleus and cytoplasm; PNS only

PNS AND CNS MYELINATION COMPARED

Feature	PNS myelination	CNS myelination
Cell that makes the myelin	Schwann cell	Oligodendrocyte
Axons per glial cell	One segment of one axon	Segments of several axons

Feature	PNS myelination	CNS myelination
Neurolemma	Present; the outer layer of the Schwann cell	Absent
Axon regeneration	Possible if the cell body survives; the neurolemma forms a regeneration tube that guides the axon	Very limited; no neurolemma, oligodendrocytes block regrowth, and astrocytes form scar tissue

MEMORY TIP

Schwann cells serve one axon segment each and leave a neurolemma. Oligodendrocytes reach several axons and leave none. That is why a cut nerve in the arm can regrow and a cut tract in the spinal cord usually cannot.

WHEN THE SUPPORT CELLS FAIL

Mature neurons generally do not divide, so tumors of nervous tissue rarely come from neurons. Most come from the **glia**, which can still divide. A **glioma** is a tumor of glial cells, and it is the most common primary malignant brain tumor.

Myelin matters as much as the axon it wraps. In **multiple sclerosis**, the immune system destroys myelin around CNS axons, and signals slow down or fail. The presence of the **neurolemma** on Schwann cells, and its absence on oligodendrocytes, is why a cut nerve in the arm can regrow while damage in the brain or spinal cord usually cannot be repaired.

The **cell body** keeps the whole neuron alive. The axon depends on the cell body for its proteins, so the part of an axon cut off from its cell body dies. An injury close to the cell body is more likely to kill the whole neuron than an injury far out along the axon.

CHAPTER 8

Gross Anatomy and Neuronal Integration

NERVOUS SYSTEM

COMPETENCIES 179 TO 183

Covers gray and white matter, the names for bundles of fibers and clusters of cell bodies, the reflex arc, types of reflexes, and neuronal pools.

COMPETENCIES

1. **179** CNS and PNS tissue terms
2. **180** Gray and white matter
3. **181** Reflex arc components
4. **182** Types of reflexes
5. **183** Neuronal pools and pathways

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Neuronal integration

drsrennie-stack.github.io/new-build-bio4-solano/nervous-tissue-concept-videos.html

COMPETENCY 179 LECTURE

CNS and PNS tissue terms

You should be able to: Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS.

Four terms name groups of neurons. Two of them name clusters of neuron cell bodies, and two name bundles of axons. Which term you use depends on whether the group sits inside the CNS or out in the PNS.

A **nucleus** is a cluster of neuron cell bodies inside the CNS. The cranial nerve nuclei in the brainstem and the basal nuclei deep in the cerebrum are examples. Nuclei are made of gray matter. This meaning is different from the nucleus inside a single cell.

A **ganglion** (plural, ganglia) is a cluster of neuron cell bodies in the PNS. The **dorsal root ganglion** holds the cell bodies of sensory neurons, and the **sympathetic trunk ganglia** hold the cell bodies of autonomic neurons. Satellite cells surround the cell bodies in a ganglion. The basal nuclei of the brain are also called the basal ganglia, an older name that breaks this rule, since they sit in the CNS.

A **tract** is a bundle of axons inside the CNS. The **corticospinal tract** and the **spinothalamic tract** are examples. Tracts make up the white matter of the brain and spinal cord.

A **nerve** is a bundle of axons in the PNS, held together by connective tissue and supplied by blood vessels. The sciatic nerve and the vagus nerve are examples. A nerve holds axons but no cell bodies. The cell bodies of those axons sit in the CNS or in ganglia.

THE SAME STRUCTURE UNDER TWO NAMES

Structure	In the CNS	In the PNS
A cluster of cell bodies	Nucleus	Ganglion
A bundle of axons	Tract	Nerve

THE FOUR TERMS, WHERE EACH ONE BELONGS, AND AN EXAMPLE

Term	Cell bodies or axons	CNS or PNS	Example
Nucleus	Cluster of cell bodies	CNS	A cranial nerve nucleus in the brainstem
Ganglion	Cluster of cell bodies	PNS	A dorsal root ganglion
Tract	Bundle of axons	CNS	The corticospinal tract
Nerve	Bundle of axons	PNS	The sciatic nerve

The wrappings of a nerve

On a nerve cross section, work from one axon outward. Each axon with its Schwann cell is wrapped in **endoneurium**. Groups of axons are bundled into a **fascicle** by **perineurium**. The whole nerve, with all its fascicles and blood vessels, is wrapped in **epineurium**.

A NERVE FROM THE INSIDE OUT

Structure	What it wraps
Endoneurium	A single axon and its Schwann cell
Perineurium	One fascicle, a bundle of axons
Fascicle	A bundle of axons inside the nerve
Epineurium	The whole nerve

MEMORY TIP

Decide first whether you are looking at cell bodies or axons. Then decide whether you are inside the CNS or outside it. Cell bodies: nucleus inside, ganglion outside. Axons: tract inside, nerve outside.

COMPETENCY 180 LECTURE AND LAB

Gray and white matter

You should be able to: Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord.

Nervous tissue in the CNS sorts into two kinds by its look. **Gray matter** is made of neuron cell bodies, dendrites, unmyelinated axons, axon terminals, and neuroglia. It is where signals are received and processed. **White matter** is made mostly of myelinated axons, with the glia that myelinate them. It carries signals from one region to another. White matter looks white because myelin is rich in lipid.

In the spinal cord

In the spinal cord, gray matter is on the inside and white matter is on the outside. On a cross section, the gray matter forms a shape like a butterfly or the letter H. Its parts are the **posterior (dorsal) horns**, the **anterior (ventral) horns**, and, in the thoracic and upper lumbar cord (T1 to L2), the **lateral horns**. The **central canal** runs through the middle of the gray matter. The white matter around it is divided into posterior, lateral, and anterior columns, and these columns hold the tracts.

In the brain

In the cerebrum, the arrangement is reversed. A thin outer layer of gray matter, the **cerebral cortex**, covers the surface, and its large neurons include the pyramidal cells. Beneath the cortex is a large mass of white matter, including the corpus callosum. Deep inside the brain, surrounded by white matter, sit islands of gray matter called deep nuclei, such as the basal nuclei of the cerebrum and the thalamus of the diencephalon.

The cerebellum has the same plan. Its outer layer, the **cerebellar cortex**, is gray matter and contains the Purkinje cells. Its inner white matter branches like a tree and is called the **arbor vitae**. In the brainstem, nuclei of gray matter are scattered among the white matter tracts.

GRAY MATTER AND WHITE MATTER COMPARED

Type	What it is made of	Where it is
Gray matter	Neuron cell bodies, dendrites, unmyelinated axons, axon terminals, and neuroglia	The cerebral and cerebellar cortex and the deep nuclei of the brain; the H-shaped inner core of the spinal cord
White matter	Myelinated axons in bundles (tracts), with their glia	Beneath the cortex of the cerebrum and cerebellum (including the arbor vitae); the outer columns of the spinal cord

WHERE THE GRAY AND WHITE MATTER SIT

Region	Gray matter	White matter
Spinal cord	Inside: posterior, anterior, and lateral horns around the central canal	Outside: posterior, lateral, and anterior columns
Cerebrum	Outside: the cerebral cortex; plus deep nuclei inside	Inside, beneath the cortex
Cerebellum	Outside: the cerebellar cortex	Inside: the arbor vitae
Brainstem	Nuclei scattered inside	Tracts around and between the nuclei

MEMORY TIP

Gray is where the cell bodies are, white is where the myelin is. The two swap places between the spinal cord and the brain, so name the organ before you name the layer.

COMPETENCY 181 LECTURE

Reflex arc components

You should be able to: Name the five components of a reflex arc in order from receptor to effector and trace a signal through them.

A **reflex** is a fast, automatic, predictable response to a stimulus. The pathway a reflex follows is the **reflex arc**. A reflex is fast because the arc is short and fixed. When the integration center is in the spinal cord, it is a spinal reflex. When it is in the brainstem, it is a cranial reflex. Either way, no conscious decision is needed, though the brain may become aware of the response afterward.

Every reflex arc has the same five components, in the same order. The signal comes in on the sensory side, is processed in the CNS, and goes out on the motor side.

1. **Receptor** detects the stimulus; it is the ending of a sensory neuron or a receptor cell next to one
2. **Sensory neuron** carries the signal to the CNS; in a spinal reflex, its cell body sits in a dorsal root ganglion, and its axon enters the spinal cord through the dorsal root
3. **Integration center** one or more synapses in the gray matter of the spinal cord or brainstem, where the signal passes to the next neuron
4. **Motor neuron** carries the command away from the CNS; in a somatic spinal reflex, its cell body sits in the anterior horn, and its axon leaves through the ventral root
5. **Effector** the muscle or gland that carries out the response

Trace a signal: the patellar reflex

The patellar (knee-jerk) reflex is the classic example. A tap on the patellar ligament, just below the kneecap, stretches the quadriceps femoris.

1. **Receptor** muscle spindles in the quadriceps femoris detect the stretch
2. **Sensory neuron** carries the signal through the femoral nerve and dorsal roots into the spinal cord at L2 to L4
3. **Integration center** the sensory neuron synapses directly on a motor neuron in the anterior horn
4. **Motor neuron** carries the command out through the ventral root and the femoral nerve
5. **Effector** the quadriceps femoris contracts, and the leg extends at the knee

Kinds of receptors

Receptors are named for the stimulus they detect.

- **Mechanoreceptors**: touch, pressure, and stretch
- **Thermoreceptors**: temperature
- **Photoreceptors**: light
- **Chemoreceptors**: chemicals
- **Nociceptors**: pain

MEMORY TIP

The arc runs in, through, and out: Receptor, Sensory neuron, Integration center, Motor neuron, Effector. The sensory neuron enters by the dorsal root, and the motor neuron leaves by the ventral root.

COMPETENCY 182 LECTURE

Types of reflexes

You should be able to: Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each.

Reflexes are sorted in three ways: by the effector, by the number of synapses in the integration center, and by whether the reflex is inborn or learned. Each pair answers a different question, so one reflex fits into all three. The patellar reflex, for example, is somatic, monosynaptic, and innate.

Somatic versus autonomic

A **somatic reflex** has skeletal muscle as its effector. The patellar reflex is a somatic reflex. An **autonomic reflex**, also called a visceral reflex, has smooth muscle, cardiac muscle, or a gland as its effector. In the **pupillary light reflex**, bright light makes the pupils constrict. The effector is the smooth muscle of the iris, and the integration center is in the midbrain.

Monosynaptic versus polysynaptic

In a **monosynaptic reflex**, the sensory neuron synapses directly on the motor neuron, so there is only one synapse in the integration center. The **stretch reflex**, such as the patellar reflex, is monosynaptic. In a **polysynaptic reflex**, one or more interneurons sit between the sensory and motor neurons, so there are two or more synapses. The **withdrawal reflex**, pulling your hand away from a hot stove, is polysynaptic. A monosynaptic reflex is the faster of the two, because each synapse adds a short delay.

Innate versus learned

An **innate reflex** is present from birth and does not have to be learned. Blinking when something touches the eye is an innate reflex. A **learned reflex**, also called an acquired reflex, is built through practice and repetition until it runs automatically. An experienced driver braking when the car ahead stops suddenly, or catching a suddenly dropped object, are learned reflexes.

THE TYPES OF REFLEXES AND WHAT EACH NAME IS BASED ON

Reflex type	Basis of the name	Example
Somatic reflex	The effector is skeletal muscle	The patellar, or knee-jerk, reflex
Autonomic reflex	The effector is smooth muscle, cardiac muscle, or a gland	The pupillary light reflex
Monosynaptic reflex	Just one synapse in the integration center	The stretch reflex, such as the patellar reflex
Polysynaptic reflex	Two or more synapses, with interneurons	The withdrawal reflex
Innate reflex	Present from birth, not learned	The blink reflex
Learned reflex	Built through practice and repetition	Braking when the car ahead stops; catching a dropped object

MEMORY TIP

Each pair asks one question. Somatic or autonomic: which effector? Mono or poly: how many synapses? Innate or learned: born with it or practiced?

COMPETENCY 183 LECTURE

Neuronal pools and pathways

You should be able to: Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation.

Neurons in the CNS work in groups. A **neuronal pool** is a group of interconnected neurons that process the same kind of information together. A pool can hold thousands of neurons. The pattern of connections inside and between pools is called a **neural circuit**, and the pattern shapes how a signal is handled.

Convergence and divergence

In a **convergence** circuit, several presynaptic neurons synapse on one postsynaptic neuron. The one neuron can then combine input from many sources. A motor neuron in the anterior horn of the spinal cord is an example. It receives input from several regions of the brain and from sensory neurons, and its single output goes to a skeletal muscle.

In a **divergence** circuit, one presynaptic neuron branches and synapses on many postsynaptic neurons. One signal spreads out to reach many cells. A single motor neuron in the brain, for example, can stimulate many motor neurons in the spinal cord, which in turn reach many muscle fibers.

Signals can also travel in series or in parallel. In **serial processing**, the signal passes along a single chain of neurons, as in a reflex arc. In **parallel processing**, the same signal travels along several pathways at the same time.

THE POOLS AND THE CIRCUIT PATTERNS

Structure	What it is
Neuronal pool	A functional group of interconnected neurons in the CNS that processes information together
Convergence	Many neurons feed their signals onto one neuron
Divergence	One neuron spreads its signal to many neurons
Serial processing	A signal passing along a single chain of neurons, as in a reflex arc

Structure	What it is
Parallel processing	A signal sent along several pathways at the same time
Integration	The combining of all the incoming signals so that a neuron produces a single response

Ascending pathways, descending pathways, and decussation

Long-distance signals travel through the CNS in tracts that run in the white matter. A tract's name usually tells you where it starts and where it ends. The spinothalamic tract runs from the spinal cord to the thalamus.

Ascending pathways carry sensory information up the spinal cord to the brain. The **spinothalamic tract**, which carries pain and temperature, is an ascending pathway. **Descending pathways** carry motor commands down from the brain to the spinal cord. The **corticospinal tract**, which runs from the cerebral cortex to the spinal cord and controls voluntary movement, is a descending pathway.

Decussation is the crossing of a pathway from one side of the CNS to the other. The main sensory and motor pathways cross, so each side of the brain senses and controls the opposite side of the body. Most corticospinal axons cross in the medulla oblongata, at the **pyramidal decussation**. Spinothalamic axons cross in the spinal cord, close to where the sensory fibers enter.

PATHWAYS AND CROSSING

Structure	What it is	Example
Ascending pathways	Tracts that carry sensory information up the spinal cord to the brain	The spinothalamic tract
Descending pathways	Tracts that carry motor commands down from the brain to the spinal cord	The corticospinal tract
Decussation	The crossing of a pathway to the opposite side of the CNS	The pyramidal decussation in the medulla oblongata

MEMORY TIP

Converge means come together onto one. Diverge means spread out from one. Ascending carries sensation up, descending carries commands down, and decussation is the crossover.

READING THE WIRING IN THE CLINIC

A tap below the kneecap triggers the **patellar reflex**, and a clinician watches the leg kick. That one test checks the whole reflex arc: the receptor, the sensory and motor neurons, and the segment of spinal cord in between. A missing reflex points to a break somewhere along that pathway.

Because the main sensory and motor pathways **decussate**, the right side of the brain senses and controls the left side of the body. This is why a stroke in one cerebral hemisphere causes weakness on the opposite side of the body.

On an MRI scan, **gray matter** and **white matter** show up as different shades because myelin is rich in lipid. That contrast lets a radiologist see the cortex, the deep nuclei, and the white matter tracts, and spot damage such as the patches of lost myelin in multiple sclerosis.

CHAPTER 9

The Spinal Cord

CENTRAL NERVOUS SYSTEM

COMPETENCIES 184 TO 186

Covers the external anatomy of the cord, its enlargements and terminations, the roots and rootlets that connect it to the spinal nerves, and the internal anatomy of its gray and white matter. The focus is on the structures and the job each one does.

COMPETENCIES

1. **184** Spinal cord external anatomy
2. **185** Roots, rootlets, and root ganglion
3. **186** Spinal cord internal anatomy

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The spinal cord

drsrennie-stack.github.io/new-build-bio4-solano/spinal-cord-concept-videos.html

COMPETENCY 184 LECTURE AND LAB

Spinal cord external anatomy

You should be able to: Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus.

The **spinal cord** is the part of the central nervous system that sits inside the vertebral canal. It is continuous with the medulla oblongata at the foramen magnum. In an adult it is about 42 to 45 cm (16 to 18 inches) long. It carries signals between the brain and the body, and it is the center for spinal reflexes.

The cord is shorter than the vertebral column. The cord stops growing in length at about 4 to 5 years of age, while the vertebral column keeps growing. So in an adult the cord ends near the disc between L1 and L2. Below that level the vertebral canal holds only nerve roots and a thin anchoring thread.

The two enlargements

The width of the cord changes along its length. It is widest in the cervical region and thinner through the thoracic region. It swells in two places, where the nerves to the limbs leave. A limb has many muscles and a large area of skin, so these segments need more neurons.

- The **cervical enlargement** is in the lower neck, from about C4 to T1. The nerves to the upper limbs (the brachial plexus) leave here.
- The **lumbar enlargement** sits behind the lower thoracic vertebrae, about T9 to T12. Texts give slightly different levels, and some place it a little lower, near T11 to L1. The nerves to the lower limbs (the lumbar and sacral plexuses) leave here.

Where the cord ends

Below the lumbar enlargement the cord tapers to a point. This cone-shaped end is the **conus medullaris**. It sits near the disc between L1 and L2.

A thin strand of pia mater runs on from the tip of the conus medullaris. This strand is the **filum terminale**. It runs down the lower vertebral canal and anchors the cord to the coccyx.

The lumbar, sacral, and coccygeal nerve roots leave the cord high up, at or above the conus. They must travel down the vertebral canal to reach their exits lower in the column. Together these long roots look like a horse's tail, so they are

called the **cauda equina**. The filum terminale runs down the middle of them.

The two grooves

Two grooves run the full length of the cord and split it into right and left halves. The **anterior median fissure** is on the front. It is deep and wide. The **posterior median sulcus** is on the back. It is shallow and narrow. On a model or a cross-section, the wide groove tells you which side is anterior.

THE ENLARGEMENTS, THE TERMINATIONS, AND THE GROOVES

Structure	Where it is	What it is
Cervical enlargement	About C4 to T1	A widening where the nerves of the upper limbs leave the cord
Lumbar enlargement	Behind the lower thoracic vertebrae, about T9 to T12 (levels vary a little by text)	A widening where the nerves of the lower limbs leave the cord
Conus medullaris	Near the disc between L1 and L2	The tapered, cone-shaped end of the spinal cord
Filum terminale	From the tip of the conus medullaris down to the coccyx	A thread of pia mater that anchors the cord to the coccyx
Cauda equina	In the vertebral canal below the conus medullaris	The bundle of lumbar, sacral, and coccygeal nerve roots running down to their exits
Anterior median fissure	Front of the cord, full length	The deep, wide groove down the front
Posterior median sulcus	Back of the cord, full length	The shallow, narrow groove down the back

MEMORY TIP

A fissure is deep and a sulcus is shallow. The deep fissure is on the front of the cord, so the anterior side is the side with the wide groove.

WHERE THE CORD ENDS, AND WHAT THAT ALLOWS

The spinal cord ends at the **conus medullaris** near the L1 to L2 disc, but the vertebral column runs much lower. Below the conus, only the **cauda equina** and the filum terminale lie in the cerebrospinal fluid. That is why a lumbar puncture is done below the end of the cord, usually between L3 and L4 or between L4 and L5. A needle can draw cerebrospinal fluid there without reaching the cord itself.

The **posterior root ganglion** holds the sensory neuron cell bodies for one spinal nerve. The varicella zoster virus (the chickenpox virus) can stay dormant in these ganglia for decades. When it reactivates as shingles, the rash follows the strip of skin served by that one spinal nerve.

A complete spinal cord injury cuts the **ascending tracts** and **descending tracts**. These tracts carry the signals between the brain and the body below the injury. So sensation and movement are lost below the level of the injury, while function above it is spared.

COMPETENCY 185 LECTURE AND LAB

Roots, rootlets, and root ganglion

You should be able to: Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons.

Each segment of the spinal cord connects to a pair of spinal nerves, one on each side. On each side, two roots join the cord to the nerve: a posterior root and an anterior root.

Each root starts as a row of small, fan-shaped bundles of axons called **rootlets**. The posterior rootlets attach in a line along the back of the cord. The anterior rootlets leave in a line along the front of the cord. The rootlets of one segment merge to form one root.

The **posterior root** (dorsal root) is the sensory root. It carries sensory axons from receptors in the body into the cord. Along the posterior root is a swelling, the **posterior root ganglion** (dorsal root ganglion). It holds the cell bodies of the sensory neurons. These are unipolar neurons. One branch of the axon runs out to a receptor and the other runs into the cord.

The **anterior root** (ventral root) is the motor root. It carries motor axons out of the cord. Somatic motor axons go to skeletal muscle. In the T1 to L2 and S2 to S4 segments, the anterior root also carries autonomic axons that lead toward smooth muscle, cardiac muscle, and glands. The cell bodies of these motor neurons sit inside the cord, in the gray matter. That is why the anterior root has no ganglion.

Just outside the cord the two roots join to form the **spinal nerve**. It holds both sensory and motor axons, so every spinal nerve is a mixed nerve. The spinal nerve leaves the vertebral canal through the **intervertebral foramen**. This opening lies between two neighboring vertebrae, formed by the notch of the vertebra above and the notch of the vertebra below. The posterior root ganglion sits in or right next to this foramen.

FROM ROOTLET TO SPINAL NERVE, AND WHAT EACH ONE CARRIES

Structure	Where it is	What it carries
Rootlets	Rows of small bundles along the back and the front of the cord	Posterior rootlets carry sensory axons in; anterior rootlets carry motor axons out
Posterior (dorsal) root	Back of the cord, between the posterior rootlets and the spinal nerve	Sensory axons only
Posterior (dorsal) root ganglion	The swelling on the posterior root, in or beside the intervertebral foramen	Cell bodies of the sensory neurons
Anterior (ventral) root	Front of the cord, between the anterior rootlets and the spinal nerve	Motor axons only
Spinal nerve	Where the two roots join, at the intervertebral foramen	Mixed: sensory and motor axons
Intervertebral foramen	The opening between two neighboring vertebrae	The exit for the spinal nerve; it holds the posterior root ganglion

A simple spinal reflex uses every one of these structures. Follow the signal in and back out.

1. **Receptor** in the skin or a muscle detects a stimulus
2. **Spinal nerve** the sensory axon travels toward the cord
3. **Posterior root ganglion** the signal passes the sensory neuron's cell body
4. **Posterior root and rootlets** the sensory axon enters the back of the cord

5. **Gray matter of the cord** the sensory axon synapses on an interneuron or a motor neuron
6. **Anterior rootlets and anterior root** the motor axon leaves the front of the cord
7. **Spinal nerve** the motor axon travels out with the sensory axons
8. **Skeletal muscle** the effector contracts

MEMORY TIP

Posterior root: sensory in, and it has a ganglion. Anterior root: motor out, and it has no ganglion. If a question puts cell bodies on a root, it is the posterior root.

COMPETENCY 186 LAB

Spinal cord internal anatomy

You should be able to: Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns.

A cross-section of the spinal cord shows **gray matter** in the center and **white matter** around it. This is the reverse of the cerebrum, where gray matter forms the outer cortex. The gray matter is shaped like a butterfly or the letter H.

Gray matter holds neuron cell bodies, dendrites, unmyelinated axons, and neuroglia. White matter is mostly myelinated axons. The fatty myelin is what makes it look white.

The gray horns

Each half of the gray matter sticks out into projections called horns. Each horn holds a different kind of neuron.

THE GRAY HORNS AND THEIR CONTENTS

Horn	Where it is	Contents
Posterior (dorsal) horn	Points toward the back of the cord; thin and narrow	Incoming sensory axons from the posterior root, and the cell bodies of interneurons
Lateral horn	A small point out to the side, between the posterior and anterior horns; only in the T1 to L2 segments	Cell bodies of autonomic (sympathetic) motor neurons
Anterior (ventral) horn	Points toward the front of the cord; wide and rounded	Cell bodies of somatic motor neurons, whose axons leave in the anterior root to skeletal muscle

The lateral horn is found only from T1 to L2 because only those segments send out sympathetic neurons. The S2 to S4 segments hold parasympathetic neurons in a similar spot in the gray matter, but they do not form a visible horn.

The central canal and the commissures

The two halves of the cord are joined across the middle by two bridges, called commissures. The **gray commissure** is the crossbar of the H. It joins the left and right gray matter. The **central canal** runs down the middle of the gray commissure. It is a tiny channel filled with cerebrospinal fluid and lined by ependymal cells. It runs the length of the cord and is continuous with the fourth ventricle above.

The **anterior white commissure** is a band of white matter just in front of the gray commissure. Here axons cross from one side of the cord to the other.

The white columns

The white matter on each side is divided into three **columns** (also called funiculi). The horns of the gray matter mark their borders.

- The **posterior column** lies between the posterior median sulcus and the posterior horn.
- The **lateral column** lies on the side, between the posterior horn and the anterior horn.
- The **anterior column** lies between the anterior horn and the anterior median fissure.

Each column is made of **tracts**. A tract is a bundle of axons in the CNS that share the same origin and destination. **Ascending tracts** carry sensory signals up to the brain. **Descending tracts** carry motor signals down from the brain. The posterior columns hold mainly ascending tracts. The lateral and anterior columns hold both ascending and descending tracts.

THE SAME TWO THINGS HAVE DIFFERENT NAMES IN THE CNS AND THE PNS

Location	Cluster of cell bodies	Bundle of axons
Central nervous system	Nucleus	Tract
Peripheral nervous system	Ganglion	Nerve

Telling the cord segments apart

Cross-sections from different levels of the cord do not look the same. The size, the shape, and the amount of gray and white matter change from top to bottom.

TELLING THE CORD SEGMENTS APART IN CROSS-SECTION

Segment	Distinguishing features
Cervical	Large diameter, oval shape, and a large amount of white matter
Thoracic	Small diameter and a small amount of gray matter; has a small lateral horn
Lumbar	Almost circular, with large anterior horns; less white matter than the cervical segments
Sacral	Relatively small, with more gray matter than white matter
Coccygeal	The smallest; looks like the lower sacral segments

White matter is thickest near the top of the cord. At the cervical level the ascending tracts from the whole body have gathered, and the descending tracts have not yet left. Lower down, fewer tracts are still running, so the white matter thins. Gray matter is largest in the enlargements, where the limbs need many motor neurons.

MEMORY TIP

To orient a slide, find the anterior median fissure first. It is the wide groove, and the wide anterior horns sit on either side of it. The thin posterior horns point toward the back of the cord, one on each side of the shallow posterior median sulcus.

CHAPTER 10

The Peripheral Nervous System

PERIPHERAL NERVOUS SYSTEM

COMPETENCIES 187 TO 189

Covers the structure of a nerve, the spinal nerves and their roots, the rami, the nerve plexuses, and the classification of sensory receptors. The cranial nerves are covered on their own sheet. The focus is on the structures and the job each one does.

COMPETENCIES

1. **187** Nerve connective tissue wrappings
2. **188** Spinal nerves and rami
3. **189** Sensory receptor classification

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The peripheral nervous system

drsrennie-stack.github.io/new-build-bio4-solano/pns-concept-videos.html

COMPETENCY 187 LECTURE AND LAB

Nerve connective tissue wrappings

You should be able to: Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps.

A **nerve** is an organ of the peripheral nervous system. It is a bundle of many axons held together by connective tissue, with blood vessels running through it. Some of its axons are myelinated and some are not. A nerve is the PNS version of a tract in the CNS.

The smallest unit in a nerve is the single **axon**, also called a nerve fiber. In the PNS, myelinated axons are wrapped by **Schwann cells**. The small gaps between neighboring Schwann cells are the **nodes of Ranvier**.

Three layers of connective tissue wrap the nerve at three levels. Work from the inside out.

- The **endoneurium** is a thin layer of loose connective tissue around each single axon, outside its Schwann cell.
- Axons are grouped into bundles called **fascicles**. The **perineurium** wraps each fascicle.
- The **epineurium** is a tough layer of dense irregular connective tissue around the whole nerve. It also fills the spaces between the fascicles and carries the nerve's blood vessels.

THE NERVE AND ITS WRAPPINGS, INSIDE TO OUTSIDE

Structure	What it is	Level it wraps
Axon	A single nerve fiber, with its Schwann cell if myelinated	Wrapped by the endoneurium
Endoneurium	The innermost wrapping, thin and delicate	One axon
Fascicle	A bundle of axons	Wrapped by the perineurium
Perineurium	The middle wrapping	One fascicle
Epineurium	The outermost wrapping, thick and tough	The whole nerve

On a nerve cross-section, the thick outer ring is the epineurium. The round clusters inside it are fascicles, each outlined by a thin ring of perineurium. The tiny circles packed inside a fascicle are axons, each surrounded by endoneurium.

MEMORY TIP

Endo means within, peri means around, and epi means upon. Skeletal muscle uses the same pattern: endomysium around one fiber, perimysium around a fascicle, epimysium around the whole muscle.

COMPETENCY 188 LECTURE AND LAB

Spinal nerves and rami

You should be able to: State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies.

Thirty one pairs of **spinal nerves** leave the spinal cord. Each one is named for the region of the vertebral column where it leaves and is numbered within that region. Every spinal nerve is a mixed nerve, carrying both sensory and motor axons.

THE 31 PAIRS OF SPINAL NERVES BY REGION

Region	Number of pairs	Names
Cervical	8	C1 to C8
Thoracic	12	T1 to T12
Lumbar	5	L1 to L5
Sacral	5	S1 to S5
Coccygeal	1	Co1

There are 8 cervical nerves but only 7 cervical vertebrae. C1 leaves between the occipital bone and the atlas. C1 to C7 each leave above the vertebra with the same number. C8 leaves between C7 and T1. From T1 down, each nerve leaves below the vertebra with the same number.

The branches of a spinal nerve

A spinal nerve is very short. Just after it passes through the intervertebral foramen, it divides into branches. The two large branches are the rami (singular, ramus). Both rami are mixed, carrying sensory and motor axons.

- The **posterior ramus** (dorsal ramus) turns toward the back. It supplies the deep muscles of the back and the skin over the back of the neck and trunk.
- The **anterior ramus** (ventral ramus) is the larger branch. It runs forward and around the body. It supplies the muscles and skin of the front and sides of the trunk and all of the upper and lower limbs. Most anterior rami join into nerve plexuses. The anterior rami of T2 to T12 are the exception and run between the ribs as the intercostal nerves.
- The **meningeal branch** turns back in through the intervertebral foramen. It supplies the vertebrae, the vertebral ligaments, the blood vessels of the cord, and the meninges.
- The **rami communicantes** are small branches that link the spinal nerve to a ganglion of the sympathetic trunk. They carry sympathetic axons.

THE BRANCHES OF A SPINAL NERVE AND THE TERRITORY EACH SUPPLIES

Branch	Territory it supplies	What it carries

Branch	Territory it supplies	What it carries
Posterior (dorsal) ramus	Deep muscles of the back and the skin of the back of the neck and trunk	Mixed: sensory and motor axons
Anterior (ventral) ramus	Muscles and skin of the front and sides of the trunk and the limbs; forms the plexuses	Mixed: sensory and motor axons
Meningeal branch	Vertebrae, vertebral ligaments, blood vessels of the cord, and the meninges	Mostly sensory axons, plus sympathetic axons to the blood vessels
White ramus communicans	Runs from the spinal nerve into a sympathetic trunk ganglion; only T1 to L2	Myelinated preganglionic sympathetic axons
Gray ramus communicans	Runs from a sympathetic trunk ganglion back to the spinal nerve; on every spinal nerve	Unmyelinated postganglionic sympathetic axons

White and gray rami communicantes

Each spinal nerve has one or more rami communicantes, and they come in two kinds. The **white ramus communicans** carries sympathetic axons from the spinal nerve into the **sympathetic trunk ganglion** (sympathetic chain ganglion). These axons are myelinated, which makes this ramus look white. White rami are found only on the T1 to L2 spinal nerves, because only those cord segments have lateral horns.

The **gray ramus communicans** carries axons the other way, from the sympathetic trunk ganglion back into the spinal nerve. These axons are unmyelinated, so this ramus looks gray. Every spinal nerve has a gray ramus. Its axons ride out in the posterior and anterior rami to the sweat glands, the arrector pili muscles, and the blood vessels of the skin and body wall in that nerve's territory.

The sympathetic trunk ganglia sit in a chain on each side of the vertebral column. Axons link them into a long cord called the **sympathetic trunk**. Competency 193 covers the trunk and the rest of the sympathetic pathway.

MEMORY TIP

White means myelinated and goes in to the ganglion; it is found only from T1 to L2. Gray means unmyelinated and comes back out to the spinal nerve; it is found on all 31 pairs.

COMPETENCY 189 LECTURE AND LAB

Sensory receptor classification

You should be able to: Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus.

A **sensory receptor** is a structure that detects a change, called a stimulus, and starts a signal in a sensory neuron. Receptors are classified in three ways: by the kind of stimulus they detect, by where they are, and by how they are built.

Every receptor gets one label from each scheme. For example, a muscle spindle is a mechanoreceptor (stimulus), a proprioceptor (location), and an encapsulated receptor (structure).

By stimulus

CLASSIFYING RECEPTORS BY THE STIMULUS THEY DETECT

Type	Stimulus it detects
Mechanoreceptors	Mechanical force, such as touch, pressure, vibration, and stretch

Type	Stimulus it detects
Thermoreceptors	Changes in temperature
Photoreceptors	Light; found in the retina of the eye
Chemoreceptors	Chemicals in solution, such as the molecules of smell and taste
Nociceptors	Stimuli that could damage tissue; they produce the sensation of pain

By location

CLASSIFYING RECEPTORS BY WHERE THEY ARE

Type	Where it is and what it monitors
Exteroceptors	At or near the body surface; they respond to stimuli from outside the body, such as touch, temperature, and the special senses
Interoceptors	Also called visceroceptors; inside the body, in the organs and blood vessels
Proprioceptors	In skeletal muscles, tendons, and joint capsules; they report the position and movement of the body

By structure

Some receptors are the bare ends of a sensory neuron's dendrites. Others have a capsule of connective tissue around the nerve ending. A third group uses a separate receptor cell that passes the signal on to a sensory neuron.

CLASSIFYING RECEPTORS BY HOW THEY ARE BUILT

Type	How it is built	Examples
Free (nonencapsulated) nerve endings	Bare dendrite tips with no capsule	Free nerve endings, tactile (Merkel) discs, and hair follicle receptors
Encapsulated nerve endings	A dendrite wrapped in a capsule of connective tissue	Tactile corpuscles, lamellar corpuscles, bulbous corpuscles, muscle spindles, and tendon organs
Specialized receptor cells	A separate cell that detects the stimulus and signals a sensory neuron	Photoreceptors of the retina, hair cells of the inner ear, and taste receptor cells

The named receptors

The receptors below are the ones to know by name. Most are in the skin or in the muscles and joints.

THE NAMED RECEPTORS: LOCATION, STIMULUS, AND CLASS

Receptor	Location	Stimulus	Class (stimulus, location, structure)
Free nerve endings	Epithelium and connective tissue throughout the body	Pain, temperature, and some touch	Nociceptor, thermoreceptor, or mechanoreceptor; exteroceptor or interoceptor; nonencapsulated
Tactile (Merkel) discs	The deepest layer of the epidermis (stratum basale)	Light touch	Mechanoreceptor; exteroceptor; nonencapsulated

Receptor	Location	Stimulus	Class (stimulus, location, structure)
Hair follicle receptors	Wrapped around the base of hair follicles	Bending of a hair (light touch)	Mechanoreceptor; exteroceptor; nonencapsulated
Tactile (Meissner) corpuscles	The dermal papillae of hairless skin, such as the fingertips, palms, soles, and lips	Light, fine (discriminative) touch	Mechanoreceptor; exteroceptor; encapsulated
Lamellar (Pacinian) corpuscles	The deep dermis and the hypodermis; also joint capsules	Deep pressure and vibration	Mechanoreceptor; exteroceptor in the skin, proprioceptor in joint capsules; encapsulated
Bulbous (Ruffini) corpuscles	The deep dermis, the hypodermis, and joint capsules	Steady deep pressure and stretching of the skin	Mechanoreceptor; exteroceptor in the skin, proprioceptor in joint capsules; encapsulated
Muscle spindles	Inside skeletal muscles, among the muscle fibers	Stretch of the muscle	Mechanoreceptor; proprioceptor; encapsulated
Tendon organs	In tendons, near where the muscle joins the tendon	Tension the muscle puts on its tendon	Mechanoreceptor; proprioceptor; encapsulated
Joint kinesthetic receptors	The capsules of synovial joints	Joint position and movement	Mechanoreceptor; proprioceptor; a mix of encapsulated and free endings

MEMORY TIP

Lamellar means layered. On a slide a lamellar corpuscle looks like a cut onion, with many layers of capsule around one dendrite, and it sits deep. A tactile corpuscle is small and sits up in a dermal papilla, close to the surface, where it senses light touch.

FOLLOWING THE NERVE TO THE DEFICIT

Because a nerve is wrapped in connective tissue, it has some protection and some ability to heal. The **epineurium** and **perineurium** hold a cut nerve's shape. A surgeon stitches these layers to line up the cut ends and repair the nerve.

The two rami serve different territories. The **anterior rami** supply the limbs and the front and sides of the trunk, so damage to the anterior rami or the plexuses they form shows up in a limb. A **posterior ramus** supplies only the deep back muscles and the skin over them, so damage there stays in the back.

Receptors are not spread evenly. The fingertips and lips are crowded with tactile corpuscles and Merkel discs, which is why those areas feel fine detail so well. The skin of the back has far fewer, so it senses touch only crudely.

CHAPTER 11

The Cranial Nerves

PERIPHERAL NERVOUS SYSTEM

COMPETENCY 190

Covers the twelve pairs of cranial nerves: the number and name of each, its classification as sensory, motor, or mixed, its function, the skull foramen it passes through, and its nuclei. The focus is on the structures and the job each one does.

COMPETENCIES 1. **190** Cranial nerves

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The cranial nerves

drsrennie-stack.github.io/new-build-bio4-solano/cranial-nerves-concept-videos.html

COMPETENCY 190 LECTURE AND LAB

Cranial nerves

You should be able to: Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen.

Twelve pairs of **cranial nerves** attach to the brain. They are part of the peripheral nervous system. They are numbered with Roman numerals, I to XII, from front to back in the order they attach to the brain. CN I and CN II attach to the forebrain. CN III to CN XII attach to the brainstem, except that most of CN XI comes from the upper cervical spinal cord.

A sentence helps with the order of the names: "On Old Olympus' Towering Tops A Finn And German Viewed Some Hops." The first letters stand for olfactory, optic, oculomotor, trochlear, trigeminal, abducens, facial, vestibulocochlear, glossopharyngeal, vagus, spinal accessory, and hypoglossal.

Sensory, motor, or mixed

Each cranial nerve is classed by the kind of axons it carries. A **sensory nerve** carries signals only into the brain. A **motor nerve** carries signals mainly out to muscles. A **mixed nerve** carries both.

- Sensory: CN I, CN II, and CN VIII. These carry the special senses of smell, vision, hearing, and balance.
- Motor: CN III, CN IV, CN VI, CN XI, and CN XII.
- Mixed: CN V, CN VII, CN IX, and CN X.

Four cranial nerves also carry parasympathetic motor axons to smooth muscle and glands: CN III, CN VII, CN IX, and CN X. For the sensory, motor, or both pattern in order, use "Some Say Marry Money But My Brother Says Big Brains Matter Most."

THE TWELVE CRANIAL NERVES: CLASS, FUNCTION, AND SKULL OPENING

Number	Name	Class	Primary function	Skull opening
CN I	Olfactory	Sensory	Smell	Olfactory foramina in the cribriform plate of the ethmoid
CN II	Optic	Sensory	Vision	Optic canal of the sphenoid
CN III	Oculomotor	Motor	Moves four of the six eye muscles, raises the upper eyelid, constricts the pupil, and focuses the lens	Superior orbital fissure

Number	Name	Class	Primary function	Skull opening
CN IV	Trochlear	Motor	Moves the superior oblique muscle of the eye; the smallest cranial nerve	Superior orbital fissure
CN V	Trigeminal	Mixed	Sensation from the face, scalp, teeth, and the mouth and nose; moves the muscles of chewing; the largest cranial nerve, with three branches	Superior orbital fissure (V1, ophthalmic), foramen rotundum (V2, maxillary), and foramen ovale (V3, mandibular)
CN VI	Abducens	Motor	Moves the lateral rectus muscle of the eye	Superior orbital fissure
CN VII	Facial	Mixed	Moves the muscles of facial expression; taste from the front two thirds of the tongue; tears from the lacrimal gland and saliva from the submandibular and sublingual glands	Internal acoustic meatus, then out through the stylomastoid foramen
CN VIII	Vestibulocochlear	Sensory	Hearing (cochlear branch) and balance (vestibular branch)	Internal acoustic meatus
CN IX	Glossopharyngeal	Mixed	Taste and general sensation from the back third of the tongue and the pharynx; helps with swallowing; saliva from the parotid gland	Jugular foramen
CN X	Vagus	Mixed	Parasympathetic supply to the thoracic and abdominal organs; sensation from those organs; muscles of the pharynx and larynx for swallowing and voice	Jugular foramen
CN XI	Accessory	Motor	Moves the sternocleidomastoid and trapezius muscles	Jugular foramen (its spinal part first enters the skull through the foramen magnum)
CN XII	Hypoglossal	Motor	Moves the tongue	Hypoglossal canal

The openings and the structures along the way

The olfactory nerve is made of many small bundles of axons. They start in the lining of the nasal cavity and pass up through the olfactory foramina in the cribriform plate. They end in the **olfactory bulb** on the underside of the frontal lobe. From the bulb, the **olfactory tract** carries the signal back to the olfactory areas of the cerebrum.

The **optic nerve** leaves the back of the eye and passes through the **optic canal**. The two optic nerves meet at the **optic chiasm**, in front of the pituitary gland. Here some axons from each eye cross to the other side. Past the chiasm, the axons continue as the **optic tract** to the thalamus.

Three openings each carry more than one nerve. The **superior orbital fissure** carries CN III, CN IV, V1, and CN VI into the orbit. The **internal acoustic meatus** carries CN VII and CN VIII into the temporal bone. The facial nerve then runs through the temporal bone and comes out at the stylomastoid foramen. The **jugular foramen** carries CN IX, CN X, and CN XI out of the skull.

WHERE EACH CRANIAL NERVE ATTACHES TO THE BRAIN

Part of the brain	Cranial nerves attached
Cerebrum (olfactory bulb)	CN I
Diencephalon (through the optic chiasm)	CN II

Part of the brain	Cranial nerves attached
Midbrain	CN III and CN IV (CN IV is the only cranial nerve that leaves the back of the brainstem)
Pons	CN V
Pontomedullary junction (where the pons meets the medulla)	CN VI, CN VII, and CN VIII
Medulla oblongata	CN IX, CN X, and CN XII
Upper cervical spinal cord, with a small part from the medulla	CN XI

The nerves to the eye muscles

Three cranial nerves move the eye and the upper eyelid. The abducens nerve supplies the lateral rectus, which abducts the eye (turns it outward). The trochlear nerve supplies the superior oblique, whose tendon runs through a pulley called the trochlea. The oculomotor nerve supplies all the rest.

THE EXTRINSIC EYE MUSCLES AND THEIR NERVES

Muscle	Nerve
Lateral rectus	CN VI, abducens
Superior oblique	CN IV, trochlear
Medial rectus	CN III, oculomotor
Superior rectus	CN III, oculomotor
Inferior rectus	CN III, oculomotor
Inferior oblique	CN III, oculomotor
Levator palpebrae superioris (raises the upper eyelid)	CN III, oculomotor

MEMORY TIP

For the eye muscles, use LR6 SO4: lateral rectus is CN VI, superior oblique is CN IV, and all the rest are CN III. For the shared openings, learn three groups: III, IV, V1, and VI through the superior orbital fissure; VII and VIII through the internal acoustic meatus; IX, X, and XI through the jugular foramen.

TWELVE NERVES, TWELVE EXAMS

Each cranial nerve has its own job, so a clinician can test the nerves one at a time. A weak or missing function points to the nerve that carries it. The nerve's attachment to the brain and its path through the skull then point to where the problem may be.

The openings group nerves together. The **jugular foramen** carries CN IX, X, and XI side by side, so a tumor or fracture there can affect all three at once. The result is a recognizable cluster of swallowing, voice, and shoulder signs.

Two common nerve problems are cranial. **Bell's palsy** is a sudden weakness of CN VII that makes one side of the face droop. **Trigeminal neuralgia** is a stabbing facial pain along the branches of CN V. The path of each nerve predicts where each one shows up.

CHAPTER 12

The Nerve Plexuses

PERIPHERAL NERVOUS SYSTEM

COMPETENCY 191

A brief reference for the nerve plexuses. A plexus is a network where the anterior rami of spinal nerves regroup before traveling to the body. Covers the principal plexuses at a general level: where each one forms and what region it serves, so you can predict the area of the body affected if a plexus is damaged.

COMPETENCIES 1. **191** Nerve plexuses

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The nerve plexuses

drsrennie-stack.github.io/new-build-bio4-solano/pns-concept-videos.html

COMPETENCY 191 LECTURE AND LAB

Nerve plexuses

You should be able to: Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged.

The anterior rami of most spinal nerves do not run straight to their targets. The anterior rami of neighboring spinal nerves first join and branch again to form a network called a **nerve plexus**. Their axons are sorted and regrouped there, and then leave the plexus in named nerves.

Because of this regrouping, each nerve that leaves a plexus carries axons from several spinal nerves. So a body region gets axons from more than one spinal nerve. Damage to one spinal nerve weakens a region but does not usually wipe out all function there.

The thoracic region is the exception. The anterior rami of T2 to T12 do not form plexuses. They run between the ribs as the **intercostal nerves** (T12, below the last rib, is called the subcostal nerve). They supply the intercostal muscles, the muscles of the abdominal wall, and the skin of the chest and abdomen.

There are five plexuses. Each is named for its region and is formed by a set range of spinal nerves.

THE FIVE PLEXUSES, WHAT FORMS EACH, AND THE REGION IT SERVES

Plexus	Formed by (anterior rami)	Main nerves	Region served
Cervical plexus	C1 to C4, with a small contribution from C5	Phrenic nerve (C3 to C5) and small nerves to the neck	The skin and muscles of the neck and upper shoulder; the phrenic nerve is the motor supply to the diaphragm
Brachial plexus	C5 to C8 and T1	Axillary, musculocutaneous, radial, median, and ulnar nerves	The shoulder and the entire upper limb
Lumbar plexus	L1 to L4	Femoral and obturator nerves	The lower abdominal wall, the external genitals, and the front and inner (medial) thigh
Sacral plexus	L4 to L5 and S1 to S4	Sciatic nerve (the largest nerve in the body), gluteal nerves, and pudendal nerve	The buttock, the perineum, the back of the thigh, and most of the leg and foot

Plexus	Formed by (anterior rami)	Main nerves	Region served
Coccygeal plexus	S4, S5, and the coccygeal nerve (Co1)	Small nerves to the skin	A small area of skin over the coccyx

MEMORY TIP

The plexuses run head to tail in the same order as the cord regions that build them: cervical for the neck and diaphragm, brachial for the upper limb, lumbar for the abdominal wall and front of the thigh, sacral for the buttock and the rest of the lower limb, and coccygeal for the skin over the coccyx.

Predicting the effect of plexus damage

Each plexus serves one region, so the plexus that is damaged predicts the region that loses movement and sensation. To answer a question, first name the plexus, then name the region it serves. The damage shows up on the same side of the body as the injured plexus.

PLEXUS DAMAGED AND THE REGION AFFECTED

Plexus damaged	Region affected
Cervical plexus	Weakness or numbness of the neck and shoulder; if the phrenic nerve is involved, the diaphragm on that side is weak and breathing is affected
Brachial plexus	Weakness and sensory loss across the shoulder and the whole upper limb on that side
Lumbar plexus	Weakness and sensory loss of the lower abdominal wall and the front and inner surfaces of the thigh
Sacral plexus	Weakness and sensory loss of the buttock, the back of the thigh, and the leg and foot
Coccygeal plexus	Numbness of the small patch of skin over the coccyx

WHY THE NETWORK MATTERS

The plexus design is a safety feature. Each limb region draws axons from several spinal nerves, so a single damaged spinal nerve root weakens an area but rarely paralyzes it completely.

Damage to the **plexus** itself is different. It can damage axons from several spinal nerves at once, so it can take out a whole region. A brachial plexus injury affects the upper limb, and a lumbar or sacral plexus injury affects the lower limb.

The **phrenic nerve** leaves the cervical plexus and is the only motor supply to the diaphragm. The phrase "C3, 4, 5 keeps the diaphragm alive" is a reminder that an injury to the cord at or above that level can stop breathing.

CHAPTER 13

The Autonomic Nervous System

AUTONOMIC NERVOUS SYSTEM

COMPETENCIES 192 TO 193

Covers the difference between the somatic and autonomic systems, the two-neuron motor pathway, the named autonomic ganglia, the sympathetic and parasympathetic divisions, and the autonomic and enteric plexuses. The focus is on the structures and the job each one does.

COMPETENCIES

1. **192** Autonomic two-neuron pathway
2. **193** Autonomic ganglia and divisions

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The autonomic nervous system

drsrennie-stack.github.io/new-build-bio4-solano/ans-concept-videos.html

COMPETENCY 192 LECTURE

Autonomic two-neuron pathway

You should be able to: Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector.

The motor side of the peripheral nervous system has two parts. The **somatic nervous system** controls skeletal muscle, and that control is usually voluntary. The **autonomic nervous system** (ANS) controls cardiac muscle, smooth muscle, and glands, and that control is involuntary. Both start in the central nervous system.

THE SOMATIC AND AUTONOMIC MOTOR SYSTEMS

Feature	Somatic nervous system	Autonomic nervous system
Effectors	Skeletal muscle	Cardiac muscle, smooth muscle, and glands
Control	Voluntary	Involuntary
Motor neurons from CNS to effector	One	Two, with a synapse in an autonomic ganglion
Where the motor cell bodies are	Anterior horn of the cord or a motor nucleus of the brainstem	First neuron in the brainstem or the cord; second neuron in an autonomic ganglion
Myelin	The axon is myelinated	The first axon is lightly myelinated; the second axon is unmyelinated
Effect on the effector	Always excites; causes contraction	Can excite or inhibit
Awareness	Normally conscious	Normally not conscious

In the somatic system one neuron does the whole job. Its cell body sits in the anterior horn of the spinal cord or in a motor nucleus of the brainstem. Its myelinated axon runs all the way out to the skeletal muscle.

The autonomic system always uses two motor neurons in a row. The first is the **preganglionic neuron**. Its cell body is in the CNS, in a brainstem nucleus or in the lateral gray matter of the spinal cord. Its axon is thin and lightly myelinated. It leaves the CNS in a cranial nerve or an anterior root and ends in an autonomic ganglion.

An **autonomic ganglion** is a cluster of neuron cell bodies outside the CNS. Here the preganglionic axon synapses with the second neuron, the **postganglionic neuron**. The postganglionic cell body sits in the ganglion. Its unmyelinated axon runs from the ganglion out to the effector.

1. **Preganglionic neuron** cell body in the brainstem or the lateral gray matter of the cord; its lightly myelinated axon leaves the CNS
2. **Autonomic ganglion** the preganglionic axon synapses with the postganglionic neuron
3. **Postganglionic neuron** cell body in the ganglion; its unmyelinated axon carries the signal the rest of the way
4. **Effector** cardiac muscle, smooth muscle, or a gland receives the signal

The ANS also has sensory neurons. They carry signals from receptors in the organs and blood vessels back to the CNS. These signals help the CNS adjust the organs.

MEMORY TIP

Pre means before the ganglion and post means after it. Do not mix up the two kinds of ganglia: a posterior root ganglion holds sensory cell bodies and has no synapses, while an autonomic ganglion holds motor cell bodies and is where the synapse happens.

COMPETENCY 193 LECTURE

Autonomic ganglia and divisions

You should be able to: Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses.

The ANS has two divisions, the sympathetic and the parasympathetic. Most organs get axons from both, and the two usually have opposite effects. The two divisions differ in where their preganglionic neurons start in the CNS and in where their ganglia sit.

The sympathetic division

The **sympathetic division** is the fight or flight division. It takes over during exercise, excitement, and emergencies. It is also called the thoracolumbar division, because its preganglionic cell bodies sit in the lateral horns of the T1 to L2 spinal segments. Its ganglia lie close to the spinal cord. So its preganglionic axons are short and its postganglionic axons are long.

The **sympathetic trunk** is a chain of ganglia on each side of the vertebral column, running from the base of the skull to the coccyx. Its ganglia are the **sympathetic trunk ganglia**, also called paravertebral or chain ganglia. Axons link one ganglion to the next. The trunk connects to the spinal nerves through the white and gray rami communicantes.

The **prevertebral ganglia**, also called collateral ganglia, lie in front of the vertebral column, close to the large branches of the abdominal aorta. They are the celiac, superior mesenteric, inferior mesenteric, aorticorenal, and renal ganglia. Preganglionic axons reach them by passing through the sympathetic trunk without synapsing, in nerves called splanchnic nerves.

Follow a sympathetic signal to a sweat gland in the skin.

1. **Lateral horn (T1 to L2)** the preganglionic cell body sends out its axon
2. **Anterior root and spinal nerve** the preganglionic axon leaves the cord
3. **White ramus communicans** the preganglionic axon enters the sympathetic trunk
4. **Sympathetic trunk ganglion** the preganglionic axon synapses with the postganglionic neuron
5. **Gray ramus communicans** the postganglionic axon returns to the spinal nerve

6 **Effector** the axon runs out in a ramus to a sweat gland, an arrector pili muscle, or a blood vessel

For the abdominal and pelvic organs, the path differs after the white ramus. The preganglionic axon passes through the trunk ganglion without synapsing and travels in a splanchnic nerve to a prevertebral ganglion. It synapses there. The postganglionic axon then follows the arteries to the organ.

The parasympathetic division

The **parasympathetic division** is the rest and digest division. It conserves energy and runs routine work such as digestion, urination, and defecation. It is also called the craniosacral division. Its preganglionic cell bodies sit in the brainstem nuclei of four cranial nerves (CN III, VII, IX, and X) and in the lateral gray matter of the S2 to S4 spinal segments.

Its ganglia are the **terminal ganglia**. They lie close to the organ or inside its wall (intramural ganglia). So its preganglionic axons are long and its postganglionic axons are short. The vagus nerve (CN X) carries preganglionic axons to terminal ganglia in the walls of the thoracic and abdominal organs. The pelvic splanchnic nerves from S2 to S4 carry them to the pelvic organs and the last part of the large intestine.

THE FOUR NAMED TERMINAL GANGLIA OF THE HEAD

Ganglion	Cranial nerve that brings the preganglionic axons	Target
Ciliary ganglion	CN III	The muscles that constrict the pupil and focus the lens
Pterygopalatine ganglion	CN VII	The lacrimal gland and the glands of the nasal cavity
Submandibular ganglion	CN VII	The submandibular and sublingual salivary glands
Otic ganglion	CN IX	The parotid salivary gland

SYMPATHETIC AND PARASYMPATHETIC DIVISIONS COMPARED

Feature	Sympathetic	Parasympathetic
Other name	Thoracolumbar division	Craniosacral division
Origin in the CNS	Lateral horns of the T1 to L2 spinal segments	Brainstem nuclei of CN III, VII, IX, and X, and the S2 to S4 spinal segments
Location of ganglia	Close to the spinal cord: sympathetic trunk ganglia and prevertebral ganglia	Close to or in the wall of the organ: terminal ganglia
Preganglionic axon	Short	Long
Postganglionic axon	Long	Short
Overall role	Fight or flight; prepares the body for action	Rest and digest; conserves energy

MEMORY TIP

The other two names map the origins. Thoracolumbar means the preganglionic neurons start from T1 to L2. Craniosacral means they start in the brainstem and in S2 to S4. The ganglia follow: sympathetic near the cord, parasympathetic at the organ.

The autonomic plexuses

In the thorax, abdomen, and pelvis, autonomic axons form networks around the large blood vessels. These are the **autonomic plexuses**. They hold sympathetic and parasympathetic axons together, along with sensory axons from the organs. Some also hold sympathetic ganglia. For example, the celiac ganglia sit inside the celiac plexus.

THE NAMED AUTONOMIC PLEXUSES

Plexus	Where it is and what it supplies
Cardiac plexus	Near the base of the heart; supplies the heart
Pulmonary plexus	Around the bronchi at the root of each lung; supplies the bronchial tree
Celiac (solar) plexus	The largest; surrounds the celiac trunk and supplies most of the upper abdominal organs
Superior and inferior mesenteric plexuses	Around the mesenteric arteries; supply the small and large intestines
Renal plexus	Around the renal arteries; supplies the kidneys and ureters
Hypogastric plexus	In front of the lower lumbar vertebrae and in the pelvis; supplies the pelvic organs

The enteric plexuses

The wall of the digestive tract has its own networks of neurons, the **enteric plexuses**. They run from the esophagus to the anus and form the enteric nervous system, which has about 100 million neurons. These networks can run the gut wall on their own, and the ANS adjusts their activity.

- The **myenteric plexus** lies in the muscularis externa, between the circular and longitudinal muscle layers. It mainly controls the movement of the gut.
- The **submucosal plexus** lies in the submucosa. It mainly controls the secretions of the gut.

TWO DIVISIONS, OPPOSITE EFFECTS

The two divisions usually pull an organ in opposite directions. So the body can adjust an organ by easing off one division as well as by pushing the other. Heart rate, for example, rises when **parasympathetic** input is withdrawn, even before the sympathetic division speeds the heart up.

The **vagus nerve**, CN X, carries most of the parasympathetic supply to the chest and abdomen. Its long reach is why one nerve can slow the heart, narrow the airways, and speed digestion.

The **enteric nervous system** is independent enough to be called the brain of the gut. Its plexuses can keep the gut wall contracting in rhythm and its glands secreting even when the connections to the CNS are cut.

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